

CHAPTER 13

Linear Operators on Inner Product Spaces

13.1 Introduction

This chapter investigates the space $A(V)$ of linear operators T on an inner product space V . (See [Chapter 7](#).) Thus, the base field K is either the real numbers $\mathbf{R}$ or the complex numbers $\mathbf{C}$. In fact, different terminologies will be used for the real case and the complex case. We also use the fact that the inner products on real Euclidean space $\mathbf{R}^n$ and complex Euclidean space $\mathbf{C}^n$ may be defined, respectively, by

$$\langle u, v \rangle = u^T v \quad \text{and} \quad \langle u, v \rangle = u^T \bar{v}$$

where u and v are column vectors.

The reader should review the material in [Chapter 7](#) and be very familiar with the notions of norm (length), orthogonality, and orthonormal bases. We also note that [Chapter 7](#) mainly dealt with real inner product spaces,

whereas here we assume that V is a complex inner product space unless otherwise stated or implied.

Lastly, we note that in [Chapter 2](#), we used A^H to denote the conjugate transpose of a complex matrix A ; that is, $A^H = \overline{A^T}$. This notation is not standard. Many texts, especially advanced texts, use A^* to denote such a matrix; we will use that notation in this chapter. That is, now $A^* = \overline{A^T}$.

13.2 Adjoint Operators

We begin with the following basic definition.

DEFINITION: A linear operator T on an inner product space V is said to have an adjoint operator T^* on V if $\langle T(u), v \rangle = \langle u, T^*(v) \rangle$ for every $u, v \in V$.

The following example shows that the adjoint operator has a simple description within the context of matrix mappings.

EXAMPLE 13.1

(a) Let A be a real n -square matrix viewed as a linear operator on $\mathbf{R}^n$. Then, for every $u, v \in \mathbf{R}^n$,

$$\langle Au, v \rangle = (Au)^T v = u^T A^T v = \langle u, A^T v \rangle$$

Thus, the transpose A^T of A is the adjoint of A .

(b) Let B be a complex n -square matrix viewed as a linear operator on $\mathbf{C}^n$. Then for every $u, v \in \mathbf{C}^n$,

$$\langle Bu, v \rangle = (Bu)^T \bar{v} = u^T B^T \bar{v} = u^T \overline{B^*} \bar{v} = \langle u, B^* v \rangle$$

Thus, the conjugate transpose B^* of B is the adjoint of B .

Remark: B^* may mean either the adjoint of B as a linear operator or the conjugate transpose of B as a matrix. By [Example 13.1\(b\)](#), the ambiguity makes no difference, because they denote the same object.

The following theorem (proved in Problem 13.4) is the main result in this section.

THEOREM 13.1: Let T be a linear operator on a finite-dimensional inner product space V over K . Then

- (i) There exists a unique linear operator T^* on V such that $\langle Tu, v \rangle = \langle u, T^*(v) \rangle$ for every $u, v \in V$. (That is, T has an adjoint T^* .)
- (ii) If A is the matrix representation of T with respect to any orthonormal basis $S = \{u_i\}$ of V , then the matrix representation of T^* in the basis S is the conjugate transpose A^* of A (or the transpose A^T of A when K is real).

We emphasize that no such simple relationship exists between the matrices representing T and T^* if the basis is not orthonormal. Thus, we see one useful property of orthonormal bases. We also emphasize that this theorem is not valid if V has infinite dimension (Problem 13.31).

The following theorem (proved in Problem 13.5) summarizes some of the properties of the adjoint.

THEOREM 13.2: Let T, T_1, T_2 be linear operators on V and let $k \in K$. Then

- (i) $(T_1 + T_2)^* = T_1^* + T_2^*$, (iii) $(T_1 T_2)^* = T_2^* T_1^*$,
- (ii) $(kT)^* = \bar{k}T^*$, (iv) $(T^*)^* = T$.

Observe the similarity between the above theorem and [Theorem 2.3](#) on properties of the transpose operation on matrices.

Linear Functionals and Inner Product Spaces

Recall ([Chapter 11](#)) that a linear functional ϕ on a vector space V is a linear mapping $\phi : V \rightarrow K$. This subsection

contains an important result ([Theorem 13.3](#)) that is used in the proof of the above basic [Theorem 13.1](#).

Let V be an inner product space. Each $u \in V$ determines a mapping $\hat{u}: V \rightarrow K$ defined by

$$\hat{u}(v) = \langle v, u \rangle$$

Now, for any $a, b \in K$ and any $v_1, v_2 \in V$,

$$\hat{u}(av_1 + bv_2) = \langle av_1 + bv_2, u \rangle = a\langle v_1, u \rangle + b\langle v_2, u \rangle = a\hat{u}(v_1) + b\hat{u}(v_2)$$

That is, $\hat{u}$ is a linear functional on V . The converse is also true for spaces of finite dimension and it is contained in the following important theorem (proved in Problem 13.3).

THEOREM 13.3: Let ϕ be a linear functional on a finite-dimensional inner product space V . Then there exists a unique vector $u \in V$ such that $\phi(v) = \langle v, u \rangle$ for every $v \in V$.

We remark that the above theorem is not valid for spaces of infinite dimension (Problem 13.24).

13.3 Analogy Between $A(V)$ and C , Special Linear Operators

Let $A(V)$ denote the algebra of all linear operators on a finite-dimensional inner product space V . The adjoint mapping $T \rightarrow T^*$ on $A(V)$ is quite analogous to the conjugation mapping $z \mapsto \bar{z}$ on the complex field C . To illustrate this analogy we identify in [Table 13-1](#) certain classes of operators $T \in A(V)$ whose behavior under the adjoint map imitates the behavior under conjugation of familiar classes of complex numbers.

The analogy between these operators T and complex numbers z is reflected in the next theorem.

Table 13-1

Class of complex numbers	Behavior under conjugation	Class of operators in $A(V)$	Behavior under the adjoint map
Unit circle ($ z = 1$)	$\bar{z} = 1/z$	Orthogonal operators (real case) Unitary operators (complex case)	$T^* = T^{-1}$
Real axis	$\bar{z} = z$	Self-adjoint operators Also called: symmetric (real case) Hermitian (complex case)	$T^* = T$
Imaginary axis	$\bar{z} = -z$	Skew-adjoint operators Also called: skew-symmetric (real case) skew-Hermitian (complex case)	$T^* = -T$
Positive real axis ($0, \infty$)	$z = \bar{w}w, w \neq 0$	Positive definite operators	$T = S^*S$ with S nonsingular

THEOREM 13.4: Let λ be an eigenvalue of a linear operator T on V .

- (i) If $T^* = T^{-1}$ (i.e., T is orthogonal or unitary), then $|\lambda| = 1$.
- (ii) If $T^* = T$ (i.e., T is self-adjoint), then λ is real.
- (iii) If $T^* = -T$ (i.e., T is skew-adjoint), then λ is pure imaginary.
- (iv) If $T = S^*S$ with S nonsingular (i.e., T is positive definite), then λ is real and positive.

Proof. In each case let v be a nonzero eigenvector of T belonging to λ that is, $T(v) = \lambda v$ with $v \neq 0$. Hence, $\langle v; v \rangle$ is positive.

Proof of (i). We show that $\lambda \bar{\lambda} \langle v, v \rangle = \langle v, v \rangle$

$$\lambda \bar{\lambda} \langle v, v \rangle = \langle \lambda v, \lambda v \rangle = \langle T(v), T(v) \rangle = \langle v, T^* T(v) \rangle = \langle v, I(v) \rangle = \langle v, v \rangle$$

But $\langle v, v \rangle \neq 0$; $\lambda \bar{\lambda} = 1$ and so $|\lambda| = 1$.

Proof of (ii). We show that $\lambda \langle v, v \rangle = \bar{\lambda} \langle v, v \rangle$

$$\lambda \langle v, v \rangle = \langle \lambda v, v \rangle = \langle T(v), v \rangle = \langle v, T^*(v) \rangle = \langle v, T(v) \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \langle v, v \rangle$$

But $\langle v, v \rangle \neq 0$; hence, $\lambda \langle v, v \rangle = \bar{\lambda} \langle v, v \rangle$ and so $|\lambda|$ is real.

Proof of (iii). We show that $\lambda \langle v, v \rangle = \lambda \langle v, v \rangle$:

$$\lambda \langle v, v \rangle = \langle \lambda v, v \rangle = \langle T(v), v \rangle = \langle v, T^*(v) \rangle = \langle v, -T(v) \rangle = \langle v, -\lambda v \rangle = -\bar{\lambda} \langle v, v \rangle$$

But $\langle v, v \rangle \neq 0$; hence, $\lambda = -\bar{\lambda}$ or $\bar{\lambda} = -\lambda$, and so λ is pure imaginary.

Proof of (iv). Note first that $S(v) \neq 0$ because S is nonsingular; hence, $\langle S(v), S(v) \rangle$ is positive. We show that $\lambda \langle v, v \rangle = \langle S(v), S(v) \rangle$:

$$\lambda \langle v, v \rangle = \langle \lambda v, v \rangle = \langle T(v), v \rangle = \langle S^* S(v), v \rangle = \langle S(v), S(v) \rangle$$

But $\langle v, v \rangle$ and $\langle S(v), S(v) \rangle$ are positive; hence, λ is positive.

Remark: Each of the above operators T commutes with its adjoint; that is, $TT^* = T^*T$. Such operators are called normal operators.

13.4 Self-Adjoint Operators

Let T be a self-adjoint operator on an inner product space V ; that is, suppose

$$T^* = T$$

(If T is defined by a matrix A , then A is symmetric or Hermitian according as A is real or complex.) By [Theorem 13.4](#), the eigenvalues of T are real. The following is another important property of T .

THEOREM 13.5: Let T be a self-adjoint operator on V . Suppose u and v are eigenvectors of T belonging to distinct eigenvalues. Then u and v are orthogonal; that is, $\langle u; v \rangle = 0$.

Proof. Suppose $T(u) = \lambda_1 u$ and $T(v) = \lambda_2 v$, where $\lambda_1 \neq \lambda_2$. We show that $\lambda_1 \langle u; v \rangle = \lambda_2 \langle u; v \rangle$:

$$\begin{aligned} \lambda_1 \langle u, v \rangle &= \langle \lambda_1 u, v \rangle = \langle T(u), v \rangle = \langle u, T^*(v) \rangle = \langle u, T(v) \rangle \\ &= \langle u, \lambda_2 v \rangle = \bar{\lambda}_2 \langle u, v \rangle = \lambda_2 \langle u, v \rangle \end{aligned}$$

(The fourth equality uses the fact that $T^* = T$, and the last equality uses the fact that the eigenvalue λ_2 is real.)

Because $\lambda_1 \neq \lambda_2$, we get $\langle u; v \rangle = 0$. Thus, the theorem is proved.

13.5 Orthogonal and Unitary Operators

Let U be a linear operator on a finite-dimensional inner product space V . Suppose

$$U^* = U^{-1} \quad \text{or equivalently} \quad UU^* = U^*U = I$$

Recall that U is said to be orthogonal or unitary according as the underlying field is real or complex. The next theorem

(proved in Problem 13.10) gives alternative characterizations of these operators.

THEOREM 13.6: The following conditions on an operator U are equivalent:

- (i) $U^* = U^{-1}$; that is, $UU^* = U^*U = I$. [U is unitary (orthogonal).]
- (ii) U preserves inner products; that is, for every $v, w \in V$, $\langle U(v), U(w) \rangle = \langle v, w \rangle$.
- (iii) U preserves lengths; that is, for every $v \in V$, $\|U(v)\| = \|v\|$.

EXAMPLE 13.2

- (a) Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear operator that rotates each vector v about the z -axis by a fixed angle θ as shown in Fig. 10-1 (Section 10.3). That is, T is defined by

$$T(x, y, z) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta, z)$$

We note that lengths (distances from the origin) are preserved under T . Thus, T is an orthogonal operator.

- (b) Let V be l_2 -space (Hilbert space), defined in Section 7.3. Let $T: V \rightarrow V$ be the linear operator defined by $T(a_1, a_2, a_3, \dots) = (0, a_1, a_2, a_3, \dots)$

$$T(a_1, a_2, a_3, \dots) = (0, a_1, a_2, a_3, \dots)$$

Clearly, T preserves inner products and lengths. However, T is not surjective, because, for example, $(1, 0, 0, \dots)$ does not belong to the image of T ; hence, T is not invertible. Thus, we see that Theorem 13.6 is not valid for spaces of infinite dimension.

An isomorphism from one inner product space into another is a bijective mapping that preserves the three basic operations of an inner product space: vector addition, scalar multiplication, and inner products. Thus, the above mappings (orthogonal and unitary) may also be characterized as the isomorphisms of V into itself. Note that such a mapping U also preserves distances, because

$$\|U(v) - U(w)\| = \|U(v - w)\| = \|v - w\|$$

Hence, U is called an *isometry*.

13.6 Orthogonal and Unitary Matrices

Let U be a linear operator on an inner product space V . By [Theorem 13.1](#), we obtain the following results.

THEOREM 13.7A: A complex matrix A represents a unitary operator U (relative to an orthonormal basis) if and only if $A^* = A^{-1}$.

THEOREM 13.7B: A real matrix A represents an orthogonal operator U (relative to an orthonormal basis) if and only if $A^T = A^{-1}$.

The above theorems motivate the following definitions (which appeared in Sections 2.10 and 2.11).

DEFINITION: A complex matrix A for which $A^* = A^{-1}$ is called a *unitary matrix*.

DEFINITION: A real matrix A for which $A^T = A^{-1}$ is called an *orthogonal matrix*.

We repeat [Theorem 2.6](#), which characterizes the above matrices.

THEOREM 13.8: The following conditions on a matrix A are equivalent:

- (i) A is unitary (orthogonal).
- (ii) The rows of A form an orthonormal set.
- (iii) The columns of A form an orthonormal set.

13.7 Change of Orthonormal Basis

Orthonormal bases play a special role in the theory of inner product spaces V . Thus, we are naturally interested in the properties of the change-of-basis matrix from one such basis to another. The following theorem (proved in Problem 13.12) holds.

THEOREM 13.9: Let $\{u_1, \dots, u_n\}$ be an orthonormal basis of an inner product space V . Then the change-of-basis matrix from $\{u_i\}$ into another orthonormal basis is unitary (orthogonal). Conversely, if $P = [a_{ij}]$ is a unitary (orthogonal) matrix, then the following is an orthonormal basis:

$$\{u'_i = a_{1i}u_1 + a_{2i}u_2 + \cdots + a_{ni}u_n : i = 1, \dots, n\}$$

Recall that matrices A and B representing the same linear operator T are similar; that is, $B = P^{-1}AP$, where P is the (nonsingular) change-of-basis matrix. On the other hand, if V is an inner product space, we are usually interested in the case when P is unitary (or orthogonal) as suggested by [Theorem 13.9](#). (Recall that P is unitary if the conjugate transpose $P^* = P^{-1}$, and P is orthogonal if the transpose $P^T = P^{-1}$.) This leads to the following definition.

DEFINITION: Complex matrices A and B are unitarily equivalent if there exists a unitary matrix P for which $B = P^*AP$. Analogously, real matrices A and B are orthogonally equivalent if there exists an orthogonal matrix P for which $B = P^TAP$.

Note that orthogonally equivalent matrices are necessarily congruent.

13.8 Positive Definite and Positive Operators

Let P be a linear operator on an inner product space V . Then

- (i) P is said to be positive definite if $P = S^*S$ for some nonsingular operators S .
- (ii) P is said to be positive (or nonnegative or semidefinite) if $P = S^*S$ for some operator S .

The following theorems give alternative characterizations of these operators.

THEOREM 13.10A: The following conditions on an operator P are equivalent:

- (i) $P = T^2$ for some nonsingular self-adjoint operator T .
- (ii) P is positive definite.
- (iii) P is self-adjoint and $\langle P(u); u \rangle > 0$ for every $u \neq 0$ in V .

The corresponding theorem for positive operators (proved in Problem 13.21) follows.

THEOREM 13.10B: The following conditions on an operator P are equivalent:

- (i) $P = T^2$ for some self-adjoint operator T .
- (ii) P is positive; that is, $P = S^*S$.
- (iii) P is self-adjoint and $\langle P(u); u \rangle \geq 0$ for every $u \in V$.

13.9 Diagonalization and Canonical Forms in Inner Product Spaces

Let T be a linear operator on a finite-dimensional inner product space V over K . Representing T by a diagonal matrix depends upon the eigenvectors and eigenvalues of T , and hence, upon the roots of the characteristic polynomial $\Delta(t)$ of T . Now $\Delta(t)$ always factors into linear polynomials over the complex field $\mathbf{C}$ but may not have any linear polynomials over the real field $\mathbf{R}$. Thus, the situation for real inner product spaces (sometimes called Euclidean spaces) is inherently different than the situation for complex inner product spaces (sometimes called unitary spaces). Thus, we treat them separately.

Real Inner Product Spaces, Symmetric and Orthogonal Operators

The following theorem (proved in Problem 13.14) holds.

THEOREM 13.11: Let T be a symmetric (self-adjoint) operator on a real finite-dimensional product space V . Then there exists an orthonormal basis of V consisting of eigenvectors of T ; that is, T can be represented by a diagonal matrix relative to an orthonormal basis.

We give the corresponding statement for matrices.

THEOREM 13.11: (Alternative Form) Let A be a real symmetric matrix. Then there exists an orthogonal matrix P such that $B = P^{-1}AP = P^TAP$ is diagonal.

We can choose the columns of the above matrix P to be normalized orthogonal eigenvectors of A ; then the diagonal entries of B are the corresponding eigenvalues.

On the other hand, an orthogonal operator T need not be symmetric, and so it may not be represented by a diagonal matrix relative to an orthonormal matrix. However, such a matrix T does have a simple canonical representation, as described in the following theorem (proved in Problem 13.16).

THEOREM 13.12: Let T be an orthogonal operator on a real inner product space V . Then there exists an orthonormal basis of V in which T is represented by a block diagonal matrix M of the form

$$M = \text{diag} \left(I_s, -I_t, \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{bmatrix}, \dots, \begin{bmatrix} \cos \theta_r & -\sin \theta_r \\ \sin \theta_r & \cos \theta_r \end{bmatrix} \right)$$

The reader may recognize that each of the 2×2 diagonal blocks represents a rotation in the corresponding two-dimensional subspace, and each diagonal entry 1 represents a reflection in the corresponding one-dimensional subspace.

Complex Inner Product Spaces, Normal and Triangular Operators

A linear operator T is said to be normal if it commutes with its adjoint—that is, if $TT^* = T^*T$. We note that normal operators include both self-adjoint and unitary operators.

Analogously, a complex matrix A is said to be *normal* if it commutes with its conjugate transpose—that is, if $AA^* = A^*A$.

EXAMPLE 13.3 Let $A = \begin{bmatrix} 1 & 1 \\ i & 3 + 2i \end{bmatrix}$. Then $A^* = \begin{bmatrix} 1 & -i \\ 1 & 3 - 2i \end{bmatrix}$

Also $AA^* = \begin{bmatrix} 2 & 3 - 3i \\ 3 + 3i & 14 \end{bmatrix} = A^*A$. Thus, A is normal.

The following theorem (proved in Problem 13.19) holds.

THEOREM 13.13: Let T be a normal operator on a complex finite-dimensional inner product space V . Then there exists an orthonormal basis of V consisting of eigenvectors of T ; that is, T can be represented by a diagonal matrix relative to an orthonormal basis.

We give the corresponding statement for matrices.

THEOREM 13.13: (Alternative Form) Let A be a normal matrix. Then there exists a unitary matrix P such that $B = P^{-1}AP = P^*AP$ is diagonal.

The following theorem (proved in Problem 13.20) shows that even nonnormal operators on unitary spaces have a relatively simple form.

THEOREM 13.14: Let T be an arbitrary operator on a complex finite-dimensional inner product space V . Then T can be represented by a triangular matrix relative to an orthonormal basis of V .

THEOREM 13.14: (Alternative Form) Let A be an arbitrary complex matrix. Then there exists a unitary matrix P such that $B = P^{-1}AP = P^*AP$ is triangular.

13.10 Spectral Theorem

The Spectral Theorem is a reformulation of the diagonalization Theorems 13.11 and 13.13.

THEOREM 13.15: (Spectral Theorem) Let T be a normal (symmetric) operator on a complex (real) finite-dimensional inner product space V . Then there exists linear operators $E_1, \dots, E_r$ on V and scalars $\lambda_1, \dots, \lambda_r$ such that

$$\begin{array}{ll} \text{(i)} & T = \lambda_1 E_1 + \lambda_2 E_2 + \dots + \lambda_r E_r, \\ \text{(ii)} & E_1 + E_2 + \dots + E_r = I, \end{array} \quad \begin{array}{ll} \text{(iii)} & E_1^2 = E_1, E_2^2 = E_2, \dots, E_r^2 = E_r, \\ \text{(iv)} & E_i E_j = 0 \text{ for } i \neq j. \end{array}$$

The above linear operators $E_1, \dots, E_r$ are projections in the sense that $E_i^2 = E_i$. Moreover, they are said to be orthogonal projections because they have the additional property that $E_i E_j = 0$ for $i \neq j$.

The following example shows the relationship between a diagonal matrix representation and the corresponding orthogonal projections.

EXAMPLE 13.4 Consider the following diagonal matrices A ; E_1, E_2, E_3 :

$$A = \begin{bmatrix} 2 & & & \\ & 3 & & \\ & & 3 & \\ & & & 5 \end{bmatrix}, \quad E_1 = \begin{bmatrix} 1 & & & \\ & 0 & & \\ & & 0 & \\ & & & 0 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 0 & & & \\ & 1 & & \\ & & 1 & \\ & & & 0 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 0 & & & \\ & 0 & & \\ & & 0 & \\ & & & 1 \end{bmatrix}$$

The reader can verify that

$$\text{(i)} \quad A = 2E_1 + 3E_2 + 5E_3, \quad \text{(ii)} \quad E_1 + E_2 + E_3 = I, \quad \text{(iii)} \quad E_i^2 = E_i, \quad \text{(iv)} \quad E_i E_j = 0 \text{ for } i \neq j.$$

SOLVED PROBLEMS

Adjoins

13.1. Find the adjoint of $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ defined by

$$F(x, y, z) = (3x + 4y - 5z, \quad 2x - 6y + 7z, \quad 5x - 9y + z)$$

First find the matrix A that represents F in the usual basis of $\mathbf{R}^3$ —that is, the matrix A whose rows are the coefficients of x ; y ; z —and then form the transpose A^T of A . This yields

$$A = \begin{bmatrix} 3 & 4 & -5 \\ 2 & -6 & 7 \\ 5 & -9 & 1 \end{bmatrix} \quad \text{and then} \quad A^T = \begin{bmatrix} 3 & 2 & 5 \\ 4 & -6 & 7 \\ -5 & 7 & 1 \end{bmatrix}$$

The adjoint F^* is represented by the transpose of A ; hence,

$$F^*(x, y, z) = (3x + 2y + 5z, 4x - 6y - 9z, -5x + 7y + z)$$

13.2. Find the adjoint of $G: \mathbf{C}^3 \rightarrow \mathbf{C}^3$ defined by

$$G(x, y, z) = [2x + (1 - i)y, (3 + 2i)x - 4iz, 2ix + (4 - 3i)y - 3z]$$

First find the matrix B that represents G in the usual basis of $\mathbf{C}^3$, and then form the conjugate transpose B^* of B . This yields

$$B = \begin{bmatrix} 2 & 1 - i & 0 \\ 3 + 2i & 0 & -4i \\ 2i & 4 - 3i & -3 \end{bmatrix} \quad \text{and then} \quad B^* = \begin{bmatrix} 2 & 3 - 2i & -2i \\ 1 + i & 0 & 4 + 3i \\ 0 & 4i & -3 \end{bmatrix}$$

Then $G^*(x; y; z) = [2x + (3 - 2i)y - 2iz; (1 + i)x + (4 + 3i)z; 4iy - 3z]$.

13.3. Prove [Theorem 13.3](#): Let ϕ be a linear functional on an n -dimensional inner product space V . Then there exists a unique vector $u \in V$ such that $\phi(v) = \langle v, u \rangle$ for every $v \in V$.

Let $\{w_1, \dots; w_n\}$ be an orthonormal basis of V . Set

$$u = \overline{\phi(w_1)}w_1 + \overline{\phi(w_2)}w_2 + \cdots + \overline{\phi(w_n)}w_n$$

Let $\hat{u}$ be the linear functional on V defined by $\hat{u}(v) = \langle v, u \rangle$ for every $v \in V$. Then, for $i = 1; \dots; n$,

$$\hat{u}(w_i) = \langle w_i, u \rangle = \langle w_i, \overline{\phi(w_1)}w_1 + \dots + \overline{\phi(w_n)}w_n \rangle = \phi(w_i)$$

Because $\hat{u}$ and ϕ agree on each basis vector, $\hat{u} = \phi$.

Now suppose u' is another vector in V for which $\phi(v) = \langle v, u' \rangle$ for every $v \in V$. Then $\langle v, u \rangle = \langle v, u' \rangle$ or $\langle v, u - u' \rangle = 0$. In particular, this is true for $v = u - u'$, and so $\langle u - u'; u - u' \rangle = 0$. This yields $u - u' = 0$ and $u = u'$. Thus, such a vector u is unique, as claimed.

13.4. Prove [Theorem 13.1](#): Let T be a linear operator on an n -dimensional inner product space V . Then

(a) There exists a unique linear operator T^* on V such that

$$\langle T(u), v \rangle = \langle u, T^*(v) \rangle \quad \text{for all } u, v \in V.$$

(b) Let A be the matrix that represents T relative to an orthonormal basis $S = \{u_i\}$. Then the conjugate transpose A^* of A represents T^* in the basis S .

(a) We first define the mapping T^* . Let v be an arbitrary but fixed element of V . The map $u \mapsto \langle T(u), v \rangle$ is a linear functional on V . Hence, by [Theorem 13.3](#), there exists a unique element $v' \in V$ such that $\langle T(u), v \rangle = \langle u, v' \rangle$ for every $u \in V$. We define $T^* : V \rightarrow V$ by $T^*(v) = v'$. Then $\langle T(u), v \rangle = \langle u, T^*(v) \rangle$ for every $u, v \in V$.

We next show that T^* is linear. For any $u, v_i \in V$ and $a, b \in K$,

$$\begin{aligned}\langle u, T^*(av_1 + bv_2) \rangle &= \langle T(u), av_1 + bv_2 \rangle = \bar{a}\langle T(u), v_1 \rangle + \bar{b}\langle T(u), v_2 \rangle \\ &= \bar{a}\langle u, T^*(v_1) \rangle + \bar{b}\langle u, T^*(v_2) \rangle = \langle u, aT^*(v_1) + bT^*(v_2) \rangle\end{aligned}$$

But this is true for every $u \in V$; hence, $T^*(av_1 + bv_2) = aT^*(v_1) + bT^*(v_2)$. Thus, T^* is linear.

(b) The matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ that represent T and T^* , respectively, relative to the orthonormal basis S are given by $a_{ij} = \langle T(u_j), u_i \rangle$ and $b_{ij} = \langle T^*(u_j), u_i \rangle$ ([Problem 13.67](#)). Hence,

$$b_{ij} = \langle T^*(u_j), u_i \rangle = \overline{\langle u_i, T^*(u_j) \rangle} = \overline{\langle T(u_i), u_j \rangle} = \bar{a}_{ji}$$

Thus, $B = A^*$, as claimed.

13.5. Prove [Theorem 13.2](#):

- (i) $(T_1 + T_2)^* = T_1^* + T_2^*$, (iii) $(T_1 T_2)^* = T_2^* T_1^*$,
(ii) $(kT)^* = \bar{k}T^*$, (iv) $(T^*)^* = T$.

(i) For any $u, v \in V$,

$$\begin{aligned}\langle (T_1 + T_2)(u), v \rangle &= \langle T_1(u) + T_2(u), v \rangle = \langle T_1(u), v \rangle + \langle T_2(u), v \rangle \\ &= \langle u, T_1^*(v) \rangle + \langle u, T_2^*(v) \rangle = \langle u, T_1^*(v) + T_2^*(v) \rangle \\ &= \langle u, (T_1^* + T_2^*)(v) \rangle\end{aligned}$$

The uniqueness of the adjoint implies

$$(T_1 + T_2)^* = T_1^* + T_2^*.$$

(ii) For any $u, v \in V$,

$$\langle (kT)(u), v \rangle = \langle kT(u), v \rangle = k\langle T(u), v \rangle = k\langle u, T^*(v) \rangle = \langle u, \bar{k}T^*(v) \rangle = \langle u, (\bar{k}T^*)(v) \rangle$$

The uniqueness of the adjoint implies $(kT)^* = \bar{k}T^*$

(iii) For any $u, v \in V$,

$$\begin{aligned}\langle (T_1 T_2)(u), v \rangle &= \langle T_1(T_2(u)), v \rangle = \langle T_2(u), T_1^*(v) \rangle \\ &= \langle u, T_2^*(T_1^*(v)) \rangle = \langle u, (T_2^* T_1^*)(v) \rangle\end{aligned}$$

The uniqueness of the adjoint implies
 $(T_1 T_2)^* = T_2^* T_1^*$.

(iv) For any $u, v \in V$,

$$\langle T^*(u), v \rangle = \overline{\langle v, T^*(u) \rangle} = \overline{\langle T(v), u \rangle} = \langle u, T(v) \rangle$$

The uniqueness of the adjoint implies $(T^*)^* = T$.

13.6. Show that (a) $I^* = I$, and (b) $\mathbf{0}^* = \mathbf{0}$.

(a) For every $u, v \in V$, $\langle I(u); v \rangle = \langle u, v \rangle = \langle u; I(v) \rangle$;
hence, $I^* = I$.

(b) For every $u, v \in V$, $\langle \mathbf{0}(u); v \rangle = \langle \mathbf{0}, v \rangle = \mathbf{0} = \langle u, \mathbf{0} \rangle = \langle u, \mathbf{0}(v) \rangle$; hence, $\mathbf{0}^* = \mathbf{0}$.

13.7. Suppose T is invertible. Show that $(T^{-1})^* = (T^*)^{-1}$.

$$I = I^* = (TT^{-1})^* = (T^{-1})^* T^*; \text{ hence, } (T^{-1})^* = (T^*)^{-1}.$$

13.8. Let T be a linear operator on V , and let W be a T -invariant subspace of V . Show that $W^\perp$ is invariant under T^* .

Let $u \in W^\perp$. If $w \in W$, then $T(w) \in W$ and so $\langle w; T^*(u) \rangle = \langle T(w); u \rangle = 0$. Thus, $T^*(u) \in W^\perp$ because it is orthogonal to every $w \in W$. Hence, $W^\perp$ is invariant under T^* .

13.9. Let T be a linear operator on V . Show that each of the following conditions implies $T = \mathbf{0}$:

(i) $\langle T(u); v \rangle = 0$ for every $u, v \in V$.

(ii) V is a complex space, and $\langle T(u); u \rangle = 0$ for every $u \in V$.

(iii) T is self-adjoint and $\langle T(u); u \rangle = 0$ for every $u \in V$.

Give an example of an operator T on a real space V for which $\langle T(u); u \rangle = 0$ for every $u \in V$ but $T \neq 0$.

[Thus, (ii) need not hold for a real space V .]

(i) Set $v = T(u)$. Then $\langle T(u); T(u) \rangle = 0$, and hence, $T(u) = 0$, for every $u \in V$. Accordingly, $T = 0$.

(ii) By hypothesis, $\langle T(v + w); v + w \rangle = 0$ for any $v, w \in V$. Expanding and setting $\langle T(v); v \rangle = 0$ and $\langle T(w); w \rangle = 0$, we find

$$\langle T(v), w \rangle + \langle T(w), v \rangle = 0 \quad (1)$$

Note w is arbitrary in (1). Substituting iw for w , and using

$\langle T(v), iw \rangle = i\langle T(v), w \rangle = -i\langle T(v), w \rangle$ and $\langle T(iw), v \rangle = \langle iT(w), v \rangle = i\langle T(w), v \rangle$, we find

$$-i\langle T(v), w \rangle + i\langle T(w), v \rangle = 0$$

Dividing through by i and adding to (1), we obtain $\langle T(w), v \rangle = 0$ for any $v, w \in V$. By (i), $T = 0$.

(iii) By (ii), the result holds for the complex case; hence we need only consider the real case.

Expanding $\langle T(v + w); v + w \rangle = 0$, we again obtain (1). Because T is self-adjoint and as it is a real space, we have $\langle T(w); v \rangle = \langle w; T(v) \rangle = \langle T(v); w \rangle$. Substituting this into (1), we obtain $\langle T(v); w \rangle = 0$ for any $v, w \in V$. By (i), $T = 0$.

For an example, consider the linear operator T on $\mathbf{R}^2$ defined by $T(x, y) = (y, x)$. Then $\langle T(u), u \rangle = 0$ for every $u \in V$, but $T \neq 0$.

Orthogonal and Unitary Operators and Matrices

13.10. Prove [Theorem 13.6](#): The following conditions on an operator U are equivalent:

- (i) $U^* = U^{-1}$; that is, U is unitary. (ii) $\langle U(v), U(w) \rangle = \langle v, w \rangle$. (iii) $\|U(v)\| = \|v\|$. Suppose (i) holds. Then, for every $v, w \in V$,

$$\langle U(v), U(w) \rangle = \langle v, U^*U(w) \rangle = \langle v, I(w) \rangle = \langle v, w \rangle$$

Thus, (i) implies (ii). Now if (ii) holds, then

$$\|U(v)\| = \sqrt{\langle U(v), U(v) \rangle} = \sqrt{\langle v, v \rangle} = \|v\|$$

Hence, (ii) implies (iii). It remains to show that (iii) implies (i).

Suppose (iii) holds. Then for every $v \in V$,

$$\langle U^*U(v), v \rangle = \langle U(v), U(v) \rangle = \langle v, v \rangle = \langle I(v), v \rangle$$

Hence, $\langle (U^*U - I)(v), v \rangle = 0$ for every $v \in V$. But $U^*U - I$ is self-adjoint (Prove!); then, by [Problem 13.9](#), we have $U^*U - I = 0$ and so $U^*U = I$. Thus, $U^* = U^{-1}$.

13.11. Let U be a unitary (orthogonal) operator on V , and let W be a subspace invariant under U . Show that $W^\perp$ is also invariant under U .

Because U is nonsingular, $U(W) = W$; that is, for any $w \in W$, there exists $\tilde{w} \in W$ such that $U(\tilde{w}) = w$. Now let $v \in W^\perp$. Then, for any $w \in W$,

$$\langle U(v), w \rangle = \langle U(v), U(w') \rangle = \langle v, w' \rangle = 0$$

Thus, $U(v)$ belongs to $W^\perp$. Therefore, $W^\perp$ is invariant under U .

13.12. Prove [Theorem 13.9](#): The change-of-basis matrix from an orthonormal basis $\{u_1, \dots, u_n\}$ into another orthonormal basis is unitary (orthogonal). Conversely, if $P = [a_{ij}]$ is a unitary (orthogonal) matrix, then the vectors $u'_j = \sum_i a_{ji} u_i$ form an orthonormal basis.

Suppose $\{v_i\}$ is another orthonormal basis and suppose

$$v_i = b_{i1}u_1 + b_{i2}u_2 + \cdots + b_{in}u_n, \quad i = 1, \dots, n \quad (1)$$

Because $\{v_i\}$ is orthonormal,

$$\delta_{ij} = \langle v_i, v_j \rangle = b_{i1}\overline{b_{j1}} + b_{i2}\overline{b_{j2}} + \cdots + b_{in}\overline{b_{jn}} \quad (2)$$

Let $B = [b_{ji}]$ be the matrix of coefficients in (1). (Then B^T is the change-of-basis matrix from $\{u_i\}$ to $\{v_i\}$.) Then $BB^* = [c_{ij}]$, where $c_{ij} = b_{i1}\overline{b_{j1}} + b_{i2}\overline{b_{j2}} + \cdots + b_{in}\overline{b_{jn}}$. By (2), $c_{ij} = \delta_{ij}$, and therefore $BB^* = I$. Accordingly, B , and hence, B^T , is unitary.

It remains to prove that $\{u'_i\}$ is orthonormal. By [Problem 13.67](#),

$$\langle u'_i, u'_j \rangle = a_{1i}\overline{a_{1j}} + a_{2i}\overline{a_{2j}} + \cdots + a_{ni}\overline{a_{nj}} = \langle C_i, C_j \rangle$$

where C_i denotes the i th column of the unitary (orthogonal) matrix $P = \frac{1}{2}a_{ij}$: Because P is unitary (orthogonal), its columns are orthonormal; hence, $\langle u'_i, u'_j \rangle = \langle C_i, C_j \rangle = \delta_{ij}$. Thus, $\{u'_i\}$ is an orthonormal basis.

Symmetric Operators and Canonical Forms in Euclidean Spaces

13.13. Let T be a symmetric operator. Show that (a) The characteristic polynomial $\Delta(t)$ of T is a product of linear polynomials (over $\mathbf{R}$); (b) T has a nonzero eigenvector.

(a) Let A be a matrix representing T relative to an orthonormal basis of V ; then $A = A^T$. Let $\Delta(t)$ be the characteristic polynomial of A . Viewing A as a complex self-adjoint operator, A has only real eigenvalues by [Theorem 13.4](#). Thus,

$$\Delta(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_n)$$

where the λ_j are all real. In other words, $\Delta(t)$ is a product of linear polynomials over $\mathbf{R}$.

(b) By (a), T has at least one (real) eigenvalue. Hence, T has a nonzero eigenvector.

13.14. Prove [Theorem 13.11](#): Let T be a symmetric operator on a real n -dimensional inner product space V . Then there exists an orthonormal basis of V consisting of eigenvectors of T . (Hence, T can be represented by a diagonal matrix relative to an orthonormal basis.)

The proof is by induction on the dimension of V . If $\dim V = 1$, the theorem trivially holds. Now suppose $\dim V = n > 1$. By [Problem 13.13](#), there exists a nonzero eigenvector v_1 of T . Let W be the space spanned by v_1 , and let u_1 be a unit vector in W , e.g., let $u_1 = v_1 / \|v_1\|$.

Because v_1 is an eigenvector of T , the subspace W of V is invariant under T . By [Problem 13.8](#), $W^\perp$ is invariant under $T^* = T$. Thus, the restriction $V = W \oplus W^\perp$ of T to $W^\perp$ is a symmetric operator. By

Theorem 7.4, $V = W \oplus W^\perp$. Hence, $\dim W^\perp = n - 1$, because $\dim W = 1$. By induction, there exists an orthonormal basis $\{u_2, \dots, u_n\}$ of $W^\perp$ consisting of eigenvectors of $V = W \oplus W^\perp$ and hence of T . But $\langle u_1, u_i \rangle = 0$ for $i = 2, \dots, n$ because $u_i \in W^\perp$. Accordingly $\{u_1, u_2, \dots, u_n\}$ is an orthonormal set and consists of eigenvectors of T . Thus, the theorem is proved.

13.15. Let $q(x, y) = 3x^2 - 6xy + 11y^2$. Find an orthonormal change of coordinates (linear substitution) that diagonalizes the quadratic form q .

Find the symmetric matrix A representing q and its characteristic polynomial $D(t)$. We have

$$A = \begin{bmatrix} 3 & -3 \\ -3 & 11 \end{bmatrix} \quad \text{and} \quad \Delta(t) = t^2 - \operatorname{tr}(A)t + |A| = t^2 - 14t + 24 = (t - 2)(t - 12)$$

The eigenvalues are $\lambda = 2$ and $\lambda = 12$. Hence, a diagonal form of q is

$$q(s, t) = 2s^2 + 12t^2$$

(where we use s and t as new variables). The corresponding orthogonal change of coordinates is obtained by finding an orthogonal set of eigenvectors of A .

Subtract $\lambda = 2$ down the diagonal of A to obtain the matrix

$$M = \begin{bmatrix} 1 & -3 \\ -3 & 9 \end{bmatrix} \quad \text{corresponding to} \quad \begin{array}{l} x - 3y = 0 \\ -3x + 9y = 0 \end{array} \quad \text{or} \quad x - 3y = 0$$

A nonzero solution is $u_1 = (3, 1)$. Next subtract $\lambda = 12$ down the diagonal of A to obtain the matrix

$$M = \begin{bmatrix} -9 & -3 \\ -3 & -1 \end{bmatrix} \quad \text{corresponding to} \quad \begin{array}{l} -9x - 3y = 0 \\ -3x - y = 0 \end{array} \quad \text{or} \quad -3x - y = 0$$

A nonzero solution is $u_2 = (-1, 3)$. Normalize u_1 and u_2 to obtain the orthonormal basis

$$\hat{u}_1 = (3/\sqrt{10}, 1/\sqrt{10}), \quad \hat{u}_2 = (-1/\sqrt{10}, 3/\sqrt{10})$$

Now let P be the matrix whose columns are $\hat{u}_1$ and $\hat{u}_2$. Then

$$P = \begin{bmatrix} 3/\sqrt{10} & -1/\sqrt{10} \\ 1/\sqrt{10} & 3/\sqrt{10} \end{bmatrix} \quad \text{and} \quad D = P^{-1}AP = P^TAP = \begin{bmatrix} 2 & 0 \\ 0 & 12 \end{bmatrix}$$

Thus, the required orthogonal change of coordinates is

$$\begin{bmatrix} x \\ y \end{bmatrix} = P \begin{bmatrix} s \\ t \end{bmatrix} \quad \text{or} \quad x = \frac{3s - t}{\sqrt{10}}, \quad y = \frac{s + 3t}{\sqrt{10}}$$

One can also express s and t in terms of x and y by using $P^{-1} = P^T$; that is,

$$s = \frac{3x + y}{\sqrt{10}}, \quad t = \frac{-x + 3y}{\sqrt{10}}$$

13.16. Prove [Theorem 13.12](#): Let T be an orthogonal operator on a real inner product space V . Then there exists an orthonormal basis of V in which T is represented by a block diagonal matrix M of the form

$$M = \text{diag} \left(1, \dots, 1, -1, \dots, -1, \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{bmatrix}, \dots, \begin{bmatrix} \cos \theta_r & -\sin \theta_r \\ \sin \theta_r & \cos \theta_r \end{bmatrix} \right)$$

Let $S = T + T^{-1} = T + T^*$. Then $S^* = (T + T^*)^* = T^* + T = S$. Thus, S is a symmetric operator on V . By [Theorem 13.11](#), there exists an orthonormal basis of V consisting of eigenvectors of S . If $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of S , then V can be decomposed into the direct sum $V = V_1 \oplus V_2 \oplus \dots \oplus V_m$ where the V_i consists of the eigenvectors of S belonging to λ_i . We claim that each V_i is invariant under T . For suppose $v \in V_i$; then $S(v) = \lambda_i v$ and

$$S(T(v)) = (T + T^{-1})T(v) = T(T + T^{-1})(v) = TS(v) = T(\lambda_i v) = \lambda_i T(v)$$

That is, $T(v) \in V_i$. Hence, V_i is invariant under T . Because the V_i are orthogonal to each other, we can restrict our investigation to the way that T acts on each individual V_i .

On a given V_i , we have $(T + T^{-1})v = S(v) = \lambda_i v$. Multiplying by T , we get

$$(T^2 - \lambda_i T + I)(v) = 0 \tag{1}$$

We consider the cases $\lambda_i = \pm 2$ and $\lambda_i \neq \pm 2$ separately. If $\lambda_i = 2$, then $(T - I)^2(v) = 0$, which leads to $(T - I)(v) = 0$ or $T(v) = v$. Thus, T restricted to this V_i is either I or $-I$.

If $\lambda_i \neq \pm 2$ then T has no eigenvectors in V_i , because, by [Theorem 13.4](#), the only eigenvalues of T are 1 or -1 . Accordingly, for $v \neq 0$, the vectors v and $T(v)$ are linearly independent. Let W be the subspace spanned by v and $T(v)$. Then W is invariant under T , because using (1) we get

$$T(T(v)) = T^2(v) = \lambda_i T(v) - v \in W$$

By [Theorem 7.4](#), $V_j = W \oplus W^\perp$. Furthermore, by [Problem 13.8](#), $W^\perp$ is also invariant under T . Thus, we can decompose V_j into the direct sum of two-dimensional subspaces W_j where the W_j are orthogonal to each other and each W_j is invariant under T . Thus, we can restrict our investigation to the way in which T acts on each individual W_j .

Because $T^2 - \lambda_i T + I = 0$, the characteristic polynomial $D(t)$ of T acting on W_j is $\Delta(t) = t^2 - \lambda_i t + 1$. Thus, the determinant of T is 1, the constant term in $\Delta(t)$. By [Theorem 2.7](#), the matrix A representing T acting on W_j relative to any orthogonal basis of W_j must be of the form

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

The union of the bases of the W_j gives an orthonormal basis of V_j , and the union of the bases of the V_j gives an orthonormal basis of V in which the matrix representing T is of the desired form.

Normal Operators and Canonical Forms in Unitary Spaces

13.17. Determine which of the following matrices is normal:

$$(a) \quad A = \begin{bmatrix} 1 & i \\ 0 & 1 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 1 & i \\ 1 & 2+i \end{bmatrix}$$

$$(a) \quad AA^* = \begin{bmatrix} 1 & i \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -i & 1 \end{bmatrix} = \begin{bmatrix} 2 & i \\ -i & 1 \end{bmatrix}, \quad A^*A = \begin{bmatrix} 1 & 0 \\ -i & 1 \end{bmatrix} \begin{bmatrix} 1 & i \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & i \\ -i & 2 \end{bmatrix}$$

Because $AA^* \neq A^*A$, the matrix A is not normal.

$$(b) \quad BB^* \begin{bmatrix} 1 & i \\ 1 & 2+i \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -i & 2-i \end{bmatrix} = \begin{bmatrix} 2 & 2+2i \\ 2-2i & 6 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -i & 2-i \end{bmatrix} \begin{bmatrix} 1 & i \\ 1 & 2+i \end{bmatrix} = B^*B$$

Because $BB^* \neq B^*B$, the matrix B is not normal.

13.18. Let T be a normal operator. Prove the following:

- (a) $T(v) = 0$ if and only if $T^*(v) = 0$. (b) $T - \lambda I$ is normal.
(c) If $T(v) = \lambda_1 v$, then $T^*(v) = \bar{\lambda}_1 v$ hence, any eigenvector of T is also an eigenvector of T^* .
(d) If $T(v) = \lambda_1 v$ and $T(w) = \lambda_2 w$ where $\lambda_1 \neq \lambda_2$, then $\lambda_1 v, w = 0$; that is, eigenvectors of T belonging to distinct eigenvalues are orthogonal.

(a) We show that $\langle T(v), T(v) \rangle = \langle T^*(v), T^*(v) \rangle$:

$$\langle T(v), T(v) \rangle = \langle v, T^*T(v) \rangle = \langle v, TT^*(v) \rangle = \langle T^*(v), T^*(v) \rangle$$

Hence, by [I]₃ in the definition of the inner product in Section 7.2, $T(v) = 0$ if and only if $T^*(v) = 0$.

(b) We show that $T - \lambda I$ commutes with its adjoint:

$$\begin{aligned} (T - \lambda I)(T - \lambda I)^* &= (T - \lambda I)(T^* - \bar{\lambda} I) = TT^* - \lambda T^* - \bar{\lambda} T + \lambda \bar{\lambda} I \\ &= T^*T - \bar{\lambda} T - \lambda T^* + \lambda \bar{\lambda} I = (T^* - \bar{\lambda} I)(T - \lambda I) \\ &= (T - \lambda I)^*(T - \lambda I) \end{aligned}$$

Thus, $T - \lambda I$ is normal.

- (c) If $T(v) = \lambda v$, then $(T - \lambda I)(v) = 0$. Now $T - \lambda I$ is normal by (b); therefore, by (a), $(T - \lambda I)^*(v) = 0$. That is, $(T^* - \bar{\lambda} I)(v) = 0$; hence, $T^*(v) = \bar{\lambda} v$.

(d) We show that $\lambda_1 \langle v, w \rangle = \lambda_2 \langle v, w \rangle$:

$$\lambda_1 \langle v, w \rangle = \langle \lambda_1 v, w \rangle = \langle T(v), w \rangle = \langle v, T^*(w) \rangle = \langle v, \bar{\lambda}_2 w \rangle = \bar{\lambda}_2 \langle v, w \rangle$$

But $\lambda_1 \neq \lambda_2$, hence, $\langle v, w \rangle = 0$.

13.19. Prove [Theorem 13.13](#): Let T be a normal operator on a complex finite-dimensional inner product space V . Then there exists an orthonormal basis of V consisting of eigenvectors of T . (Thus, T can be represented by a diagonal matrix relative to an orthonormal basis.)

The proof is by induction on the dimension of V . If $\dim V = 1$, then the theorem trivially holds. Now suppose $\dim V = n > 1$. Because V is a complex vector space, T has at least one eigenvalue and hence a nonzero eigenvector v . Let W be the subspace of V spanned by v , and let u_1 be a unit vector in W .

Because v is an eigenvector of T , the subspace W is invariant under T . However, v is also an eigenvector of T^* by [Problem 13.18](#); hence, W is also invariant under T^* . By [Problem 13.8](#), $W^\perp$ is invariant under $T^{**} = T$. The remainder of the proof is identical with the latter part of the proof of [Theorem 13.11](#) ([Problem 13.14](#)).

13.20. Prove [Theorem 13.14](#): Let T be any operator on a complex finite-dimensional inner product space V . Then T can be represented by a triangular matrix relative to an orthonormal basis of V .

The proof is by induction on the dimension of V . If $\dim V = 1$, then the theorem trivially holds. Now suppose $\dim V = n > 1$. Because V is a complex vector space, T has at least one eigenvalue and hence at least one nonzero eigenvector v . Let W be the subspace of V spanned by v , and let u_1 be a unit vector in W . Then u_1 is an eigenvector of T and, say, $T(u_1) = a_{11}u_1$.

By [Theorem 7.4](#), $V = W \oplus W^\perp$. Let E denote the orthogonal projection V into $W^\perp$. Clearly $W^\perp$ is invariant under the operator ET . By induction, there exists an orthonormal basis $\{u_2, \dots; u_n\}$ of $W^\perp$ such that, for $i = 2; \dots; n$,

$$ET(u_i) = a_{i2}u_2 + a_{i3}u_3 + \dots + a_{ii}u_i$$

(Note that $u_1, u_2, \dots; u_n$ is an orthonormal basis of V .) But E is the orthogonal projection of V onto $W^\perp$; hence, we must have

$$T(u_i) = a_{i1}u_1 + a_{i2}u_2 + \dots + a_{ii}u_i$$

for $i = 2; \dots; n$. This with $T(u_1) = a_{11}u_1$ gives us the desired result.

Miscellaneous Problems

13.21.. Prove [Theorem 13.10B](#): The following are equivalent:

- (i) $P = T^2$ for some self-adjoint operator T .
- (ii) $P = S^*S$ for some operator S ; that is, P is positive.
- (iii) P is self-adjoint and $\langle P(u), u \rangle \geq 0$ for every $u \in V$.

Suppose (i) holds; that is, $P = T^2$ where $T = T^*$. Then $P = TT = T^*T$, and so (i) implies (ii). Now suppose (ii) holds. Then $P^* = (S^*S)^* = S^*S^{**} = S^*S = P$, and so P is self-adjoint. Furthermore,

$$\langle P(u), u \rangle = \langle S^*S(u), u \rangle = \langle S(u), S(u) \rangle \geq 0$$

Thus, (ii) implies (iii), and so it remains to prove that (iii) implies (i).

Now suppose (iii) holds. Because P is self-adjoint, there exists an orthonormal basis $\{u_1, \dots, u_n\}$ of V consisting of eigenvectors of P ; say, $P(u_j) = \lambda_j u_j$. By [Theorem 13.4](#), the λ_j are real. Using (iii), we show that the λ_j are nonnegative. We have, for each i ,

$$0 \leq \langle P(u_i), u_i \rangle = \langle \lambda_i u_i, u_i \rangle = \lambda_i \langle u_i, u_i \rangle$$

Thus, $\langle u_i, u_i \rangle$ defined by 0 forces $\lambda_i \geq 0$; as claimed. Accordingly, $\sqrt{\lambda_i}$ is a real number. Let T be the linear operator

$$T(u_i) = \sqrt{\lambda_i} u_i \quad \text{for } i = 1, \dots, n$$

Because T is represented by a real diagonal matrix relative to the orthonormal basis $\{u_j\}$, T is self-adjoint. Moreover, for each i ,

$$T^2(u_i) = T(\sqrt{\lambda_i} u_i) = \sqrt{\lambda_i} T(u_i) = \sqrt{\lambda_i} \sqrt{\lambda_i} u_i = \lambda_i u_i = P(u_i)$$

Because T^2 and P agree on a basis of V ; $P = T^2$. Thus, the theorem is proved.

Remark: The above operator T is the unique positive operator such that $P = T^2$; it is called the *positive square root* of P .

13.22. Show that any operator T is the sum of a self-adjoint operator and a skew-adjoint operator.

Set $S = \frac{1}{2}(T + T^*)$ and $U = \frac{1}{2}(T - T^*)$. Then $T = S + U$, where

$$\begin{aligned} \text{and } S^* &= \left[\frac{1}{2}(T + T^*) \right]^* = \frac{1}{2}(T^* + T^{**}) = \frac{1}{2}(T^* + T) = S \\ U^* &= \left[\frac{1}{2}(T - T^*) \right]^* = \frac{1}{2}(T^* - T) = -\frac{1}{2}(T - T^*) = -U \end{aligned}$$

that is, S is self-adjoint and U is skew-adjoint.

13.23. Prove: Let T be an arbitrary linear operator on a finite-dimensional inner product space V . Then T is a product of a unitary (orthogonal) operator U and a unique positive operator P ; that is, $T = UP$. Furthermore, if T is invertible, then U is also uniquely determined.

By [Theorem 13.10](#), T^*T is a positive operator; hence, there exists a (unique) positive operator P such that $P^2 = T^*T$ ([Problem 13.43](#)). Observe that

$$\|P(v)\|^2 = \langle P(v), P(v) \rangle = \langle P^2(v), v \rangle = \langle T^*T(v), v \rangle = \langle T(v), T(v) \rangle = \|T(v)\|^2 \quad (1)$$

We now consider separately the cases when T is invertible and noninvertible.

If T is invertible, then we set $\hat{U} = PT^{-1}$. We show that $\hat{U}$ is unitary:

$$\hat{U}^* = (PT^{-1})^* = T^{-1*}P^* = (T^*)^{-1}P \quad \text{and} \quad \hat{U}^*\hat{U} = (T^*)^{-1}PPT^{-1} = (T^*)^{-1}T^*TT^{-1} = I$$

Then $\hat{U}$ is also unitary, and $T = UP$ as required.

To prove uniqueness, we assume $T = U_0P_0$, where U_0 is unitary and P_0 is positive. Then

$$T^*T = P_0^*U_0^*U_0P_0 = P_0IP_0 = P_0^2$$

But the positive square root of T^*T is unique ([Problem 13.43](#)); hence, $P_0 = P$. (Note that the invertibility of T is not used to prove the uniqueness of P .) Now if T is invertible, then P is also invertible by (1). Multiplying $U_0P = UP$ on the right by P^{-1} yields $U_0 = U$. Thus, U is also unique when T is invertible.

Now suppose T is not invertible. Let W be the image of P that is, $W = \text{Im } P$. We define $U_1: W \rightarrow V$ by

$$U_1(w) = T(v), \quad \text{where} \quad P(v) = w \quad (2)$$

We must show that U_1 is well defined; that is, that $P(v) = P(v')$ implies $T(v) = T(v')$. This follows from the fact that $P(v - v') = 0$ is equivalent to $\|P(v - v')\| = 0$, which forces $\|T(v - v')\| = 0$ by (1). Thus, U_1 is well defined. We next define $U_2: W^\perp \rightarrow V$. Note that, by (1), P and T have the same kernels. Hence, the images of P and T have the same dimension; that is, $\dim(\text{Im } P) = \dim W = \dim(\text{Im } T)$. Consequently, $W^\perp$ and $(\text{Im } T)^\perp$ also have the same dimension. We let U_2 be any isomorphism between $W^\perp$ and $(\text{Im } T)^\perp$.

We next set $U = U_1 \oplus U_2$. [Here U is defined as follows: If $v \in V$ and $v = w + w'$, where $w \in W$, $w' \in W^\perp$, then $U(v) = U_1(w) + U_2(w')$.] Now U is linear ([Problem 13.69](#)), and, if $v \in V$ and $P(v) = w$, then, by (2),

$$T(v) = U_1(w) = U(w) = UP(v)$$

Thus, $T = UP$, as required.

It remains to show that U is unitary. Now every vector $x \in V$ can be written in the form $x = P(v) + w'$, where $w' \in W^\perp$. Then $U(x) = UP(v) + U_2(w') = T(v) + U_2(w')$, where $\langle T(v), U_2(w') \rangle = 0$ by definition

of U_2 . Also, $\langle T(v), T(v) \rangle = \langle P(v), P(v) \rangle$ by (1). Thus,

$$\begin{aligned} \langle U(x), U(x) \rangle &= \langle T(v) + U_2(w'), T(v) + U_2(w') \rangle = \langle T(v), T(v) \rangle + \langle U_2(w'), U_2(w') \rangle \\ &= \langle P(v), P(v) \rangle + \langle w', w' \rangle = \langle P(v) + w', P(v) + w' \rangle = \langle x, x \rangle \end{aligned}$$

[We also used the fact that $\sum_{i=1}^n w_i = 0$: Thus, U is unitary, and the theorem is proved.]

13.24. Let V be the vector space of polynomials over $\mathbf{R}$ with inner product defined by

$$\langle f, g \rangle = \int_0^1 f(t)g(t) dt$$

Give an example of a linear functional f on V for which [Theorem 13.3](#) does not hold—that is, for which there is no polynomial $h(t)$ such that $\phi(f) = \langle f, h \rangle$ for every $f \in V$.

Let $\phi: V \rightarrow \mathbf{R}$ be defined by $\phi(f) = f(0)$; that is, ϕ evaluates $f(t)$ at 0, and hence maps $f(t)$ into its constant term. Suppose a polynomial $h(t)$ exists for which

$$\phi(f) = f(0) = \int_0^1 f(t)h(t) dt \tag{1}$$

for every polynomial $f(t)$. Observe that ϕ maps the polynomial $tf(t)$ into 0; hence, by (1),

$$\int_0^1 tf(t)h(t) dt = 0 \tag{2}$$

for every polynomial $f(t)$. In particular (2) must hold for $f(t) = th(t)$; that is,

$$\int_0^1 t^2 h^2(t) dt = 0$$

This integral forces $h(t)$ to be the zero polynomial; hence, $\phi(f) = \langle f, h \rangle = \langle f, 0 \rangle = 0$ for every polynomial $f(t)$. This contradicts the fact that f is not the zero functional; hence, the polynomial $h(t)$ does not exist.

ANSWERS TO SUPPLEMENTARY PROBLEMS

Adjoint Operators

13.25. Find the adjoint of:

$$(a) \quad A = \begin{bmatrix} 5 - 2i & 3 + 7i \\ 4 - 6i & 8 + 3i \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 3 & 5i \\ i & -2i \end{bmatrix}, \quad (c) \quad C = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

13.26. Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be defined by $T(x, y, z) = (x + 2y, 3x, 4z, y)$: Find $T^*(x, y, z)$:

13.27. Let $T: \mathbf{C}^3 \rightarrow \mathbf{C}^3$ be defined by $T(x, y, z) = [ix + (2 + 3i)y; 3x + (3 - i)z; (2 - 5i)y + iz]$: Find $T^*(x, y, z)$:

13.28. For each linear function f on V ; find $u \in V$ such that $f(v) = \langle v, u \rangle$ for every $v \in V$:

(a) $\phi: \mathbf{R}^3 \rightarrow \mathbf{R}$ defined by $\phi(x, y, z) = x + 2y - 3z$:

(b) $\phi: \mathbf{C}^3 \rightarrow \mathbf{C}$ defined by $\phi(x, y, z) = ix + (2 + 3i)y + (1 - 2i)z$:

13.29. Suppose V has finite dimension. Prove that the image of T^* is the orthogonal complement of the kernel of T ; that is, $\text{Im } T^* = (\text{Ker } T)^\perp$: Hence, $\text{rank}(T) = \text{rank}(T^*)$.

13.30. Show that $T^*T = 0$ implies $T = 0$.

13.31. Let V be the vector space of polynomials over $\mathbf{R}$ with inner product defined by $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$. Let $\mathbf{D}$

be the derivative operator on V ; that is, $\mathbf{D}(f) = df/dt$. Show that there is no operator $\mathbf{D}^*$ on V such that $\langle \mathbf{D}(f), g \rangle = \langle f, \mathbf{D}^*(g) \rangle$ for every $f, g \in V$. That is, $\mathbf{D}$ has no adjoint.

Unitary and Orthogonal Operators and Matrices

13.32. Find a unitary (orthogonal) matrix whose first row is

(a) $(2/\sqrt{13}, 3/\sqrt{13})$, (b) a multiple of $(1; 1 - i)$, (c) a multiple of $(1, -i \ 1 - i)$

13.33. Prove that the products and inverses of orthogonal matrices are orthogonal. (Thus, the orthogonal matrices form a group under multiplication, called the *orthogonal group*.)

13.34. Prove that the products and inverses of unitary matrices are unitary. (Thus, the unitary matrices form a group under multiplication, called the *unitary group*.)

13.35. Show that if an orthogonal (unitary) matrix is triangular, then it is diagonal.

13.36. Recall that the complex matrices A and B are unitarily equivalent if there exists a unitary matrix P such that $B = P^*AP$. Show that this relation is an equivalence relation.

13.37. Recall that the real matrices A and B are orthogonally equivalent if there exists an orthogonal matrix P such that $B = P^TAP$. Show that this relation is an equivalence relation.

13.38. Let W be a subspace of V . For any $v \in V$, let $v = w + w'$, where $w \in W$, $w' \in W^\perp$. (Such a sum is unique because $V = W \oplus W^\perp$.) Let $T: V \rightarrow V$ be defined by $T(v)$

$= w - w'$. Show that T is self-adjoint unitary operator on V .

13.39. Let V be an inner product space, and suppose $U: V \rightarrow V$ (not assumed linear) is surjective (onto) and preserves inner products; that is, $\langle U(v); U(w) \rangle = \langle v, w \rangle$ for every $v, w \in V$. Prove that U is linear and hence unitary.

Positive and Positive Definite Operators

13.40. Show that the sum of two positive (positive definite) operators is positive (positive definite).

13.41. Let T be a linear operator on V and let $f: V \times V \rightarrow K$ be defined by $f(u, v) = \langle T(u), v \rangle$. Show that f is an inner product on V if and only if T is positive definite.

13.42. Suppose E is an orthogonal projection onto some subspace W of V . Prove that $kI + E$ is positive (positive definite) if $k \geq 0$ ($k > 0$).

13.43. Consider the operator T defined by $T(u_i) = \sqrt{\lambda_i} u_i, i = 1, \dots, n$, in the proof of [Theorem 13.10A](#). Show that T is positive and that it is the only positive operator for which $T^2 = P$

13.44. Suppose P is both positive and unitary. Prove that $P = I$.

13.45. Determine which of the following matrices are positive (positive definite):

$$(i) \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, (ii) \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}, (iii) \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, (iv) \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, (v) \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

$$(vi) \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

13.46. Prove that a 2×2 complex matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is positive if and only if (i) $A = A^*$, and (ii) a, d and $|A| = ad - bc$ are nonnegative real numbers.

13.47. Prove that a diagonal matrix A is positive (positive definite) if and only if every diagonal entry is a nonnegative (positive) real number.

Self-adjoint and Symmetric Matrices

13.48. For any operator T , show that $T + T^*$ is self-adjoint and $T - T^*$ is skew-adjoint.

13.49. Suppose T is self-adjoint. Show that $T^2(v) = 0$ implies $T(v) = 0$. Using this to prove that $T^n(v) = 0$ also implies that $T(v) = 0$ for $n > 0$.

13.50. Let V be a complex inner product space. Suppose $T(v), v$ is real for every $v \in V$. Show that T is self-adjoint.

13.51. Suppose T_1 and T_2 are self-adjoint. Show that $T_1 T_2$ is self-adjoint if and only if T_1 and T_2 commute; that is, $T_1 T_2 = T_2 T_1$.

13.52. For each of the following symmetric matrices A , find an orthogonal matrix P and a diagonal matrix D such that $P^T A P$ is diagonal:

$$(a) \ A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}, (b) \ A = \begin{bmatrix} 5 & 4 \\ 4 & -1 \end{bmatrix}, (c) \ A = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}$$

13.53. Find an orthogonal change of coordinates $X = PX$ that diagonalizes each of the following quadratic forms and find the corresponding diagonal quadratic form $q(x)$:

- (a) $q(x, y) = 2x^2 - 6xy + 10y^2$, (b) $q(x, y) = x^2 + 8xy - 5y^2$
(c) $q(x, y, z) = 2x^2 - 4xy + 5y^2 + 2xz - 4yz + 2z^2$

Normal Operators and Matrices

- 13.54.** Let $A = \begin{bmatrix} 2 & i \\ i & 2 \end{bmatrix}$. Verify that A is normal. Find a unitary matrix P such that P^*AP is diagonal. Find P^*AP .
- 13.55.** Show that a triangular matrix is normal if and only if it is diagonal.
- 13.56.** Prove that if T is normal on V , then $\|T(v)\| = \|T^*(v)\|$ for every $v \in V$. Prove that the converse holds in complex inner product spaces.
- 13.57.** Show that self-adjoint, skew-adjoint, and unitary (orthogonal) operators are normal.
- 13.58.** Suppose T is normal. Prove that
- (a) T is self-adjoint if and only if its eigenvalues are real.
 - (b) T is unitary if and only if its eigenvalues have absolute value 1.
 - (c) T is positive if and only if its eigenvalues are nonnegative real numbers.
- 13.59.** Show that if T is normal, then T and T^* have the same kernel and the same image.
- 13.60.** Suppose T_1 and T_2 are normal and commute. Show that $T_1 + T_2$ and $T_1 T_2$ are also normal.
- 13.61.** Suppose T_1 is normal and commutes with T_2 . Show that T_1 also commutes with T_2^* .

- 13.62.** Prove the following: Let T_1 and T_2 be normal operators on a complex finite-dimensional vector space V . Then there exists an orthonormal basis of V consisting of eigenvectors of both T_1 and T_2 . (That is, T_1 and T_2 can be simultaneously diagonalized.)

Isomorphism Problems for Inner Product Spaces

- 13.63.** Let $S = \{u_1, \dots, u_n\}$ be an orthonormal basis of an inner product space V over K . Show that the mapping $v \mapsto [v]_S$ is an (inner product space) isomorphism between V and K^n . (Here $[v]_S$ denotes the coordinate vector of v in the basis S .)
- 13.64.** Show that inner product spaces V and W over K are isomorphic if and only if V and W have the same dimension.
- 13.65.** Suppose $\{u_1, \dots, u_n\}$ and $\{u'_1, \dots, u'_n\}$ are orthonormal bases of V and W , respectively. Let $T: V \rightarrow W$ be the linear map defined by $T(u_i) = u'_i$ for each i . Show that T is an isomorphism.
- 13.66.** Let V be an inner product space. Recall that each $u \in V$ determines a linear functional $\hat{u}$ in the dual space V^* by the definition $(v) = \langle v, u \rangle$ for every $v \in V$. (See the text immediately preceding [Theorem 13.3](#).) Show that the map $u \mapsto \hat{u}$ is linear and nonsingular, and hence an isomorphism from V onto V^* .

Miscellaneous Problems

- 13.67.** Suppose $\{u_1, \dots, u_n\}$ is an orthonormal basis of V . Prove

(a)

(a) $\langle a_1u_1 + a_2u_2 + \cdots + a_nu_n, b_1u_1 + b_2u_2 + \cdots + b_nu_n \rangle = a_1\bar{b}_1 + a_2\bar{b}_2 + \cdots + a_n\bar{b}_n$

(b) Let $A = [a]_{ij}$ be the matrix representing $T: V \rightarrow V$ in the basis $\{u_i\}$: Then $a_{ij} = \langle T(u_i); u_j \rangle$:

13.68. Show that there exists an orthonormal basis $\{u_1, \dots; u_n\}$ of V consisting of eigenvectors of T if and only if there exist orthogonal projections $E_1, \dots; E_r$ and scalars $\lambda_1, \dots; \lambda_r$ such that

(i) $T = \lambda_1 E_1 + \cdots + \lambda_r E_r$, (ii) $E_1 + \cdots + E_r = I$, (iii) $E_i E_j = 0$ for $i \neq j$

13.69. Suppose $V = U \oplus W$ and suppose $T_1: U \rightarrow V$ and $T_2: W \rightarrow V$ are linear. Show that $T = T_1 T_2$ is also linear. Here T is defined as follows: If $v \in V$ and $v = u + w$ where $u \in U, w \in W$, then

$$T(v) = T_1(u) + T_2(w)$$

ANSWERS TO SUPPLEMENTARY PROBLEMS

Notation: $[R_1, R_2, \dots; R_n]$ denotes a matrix with rows $R_1, R_2, \dots; R_n$.

13.25. (a) $[5 + 2i, 4 + 6i, 3 - 7i, 8 - 3i]$, (b) $[3, i, 5i, 2i]$, (c) $[1, 2; 1, 3]$

13.26. $T^*(x, y, z) = (x + 3y, 2x + z, 4y)$

13.27. $T^*(x, y, z) = [-ix + 3y, (2 - 3i)x + (2 + 5i)z; (3 + i)y - iz]$

13.28. (a) $u = (1, 2, 3)$, (b) $u = (-i, 2, 3i, 1 + 2i)$

13.32. (a) $(1/\sqrt{13})[2, 3; 3, -2]$, (b) $(1/\sqrt{3})[1, 1-i; 1+i, -1]$,
(c) $\frac{1}{2}[1, -i, 1-i; \sqrt{2}i, -\sqrt{2}, 0; 1, -i, -1+i]$

13.45. Only (i) and (v) are positive. Only (v) is positive definite.

13.52. (a and b) $P = (1/\sqrt{5})[2, -1; 1, 2]$,
(c) $P = (1/\sqrt{10})[3, -1; 1, 3]$ (a) $D = [2, 0, 0; -3]$, (b) $D = [7, 0; 0, -3]$, (c) $D = [8, 0; 0, -2]$

13.53. (a) $x = (3x' - y')/\sqrt{10}$, $y = (x' + 3y')/\sqrt{10}$;
(b) $x = (2x' - y')/\sqrt{5}$, $y = (x' + 2y')/\sqrt{5}$;
(c) $x = x'/\sqrt{3} + y'/\sqrt{2} + z'/\sqrt{6}$, $y = x'/\sqrt{3} - 2z'/\sqrt{6}$, $z = x'/\sqrt{3} - y'/\sqrt{2} + z'/\sqrt{6}$;
(a) $q(x') = \text{dia}(1 \ 11)$; (b) $q(x') = \text{dia}(3, -7)$; (c) $q(x') = \text{dia}(1, 17)$

13.54. (a) $P = (1/\sqrt{2})[1, -1; 1, 1]$, $P^*AP = \text{diag}(2+i, 2-i)$

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APPENDIX A

Multilinear Products

A.1 Introduction

The material in this appendix is much more abstract than that which has previously appeared. Accordingly, many of the proofs will be omitted. Also, we motivate the material with the following observation.

Let S be a basis of a vector space V . [Theorem 5.2](#) may be restated as follows.

THEOREM 5.2: Let $g : S \rightarrow V$ be the inclusion map of the basis S into V . Then, for any vector space U and any mapping $f : S \rightarrow U$; there exists a unique linear mapping $f^* : V \rightarrow U$ such that $f = f^* \cdot g$.

Another way to state the fact that $f = f^* \cdot g$ is that the diagram in [Fig. A-1\(a\)](#) *commutes*.

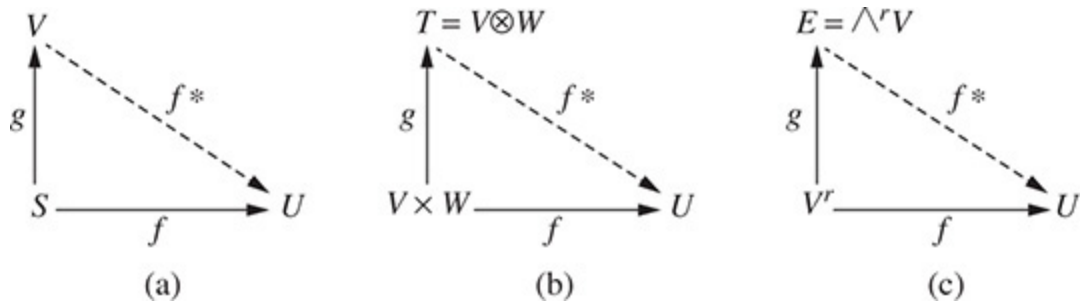


Figure A-1

A.2 Bilinear Mapping and Tensor Products

Let U, V, W be vector spaces over a field $\mathbf{K}$. Consider a map

$$f: V \times W \rightarrow U$$

Then f is said to be *bilinear* if, for each $v \in V$; the map $f_v: W \rightarrow U$ defined by $f_v(w) = f(v, w)$ is linear; and, for each $w \in W$; the map $f_w: V \rightarrow U$ defined by $f_w(v) = f(v, w)$ is linear.

That is, f is linear in each of its two variables. Note that f is similar to a bilinear form except that the values of the map f are in a vector space U rather than the field $\mathbf{K}$.

DEFINITION A.1: Let V and W be vector spaces over the same field $\mathbf{K}$. The tensor product of V and W is a vector space T over $\mathbf{K}$ together with a bilinear map $g: V \times W \rightarrow T$; denoted by $g(v, w) = v \otimes w$; with the following property: (*) For any vector space U over $\mathbf{K}$ and any bilinear map $f: V \times W \rightarrow U$ there exists a unique linear map $f^*: T \rightarrow U$ such that $f^* \circ g = f$.

The tensor product (T, g) [or simply T when g is understood] of V and W is denoted by $V \otimes W$; and the

element $v \otimes w$ is called the tensor of v and w .

Another way to state condition (*) is that the diagram in Fig. A-1(b) commutes. The fact that such a unique linear map f^* exists is called the “Universal Mapping Principle” (UMP). As illustrated in Fig. A-1(b), condition (*) also says that any bilinear map $f: V \times W \rightarrow U$ “factors through” the tensor product $T = V \otimes W$: The uniqueness in (*) implies that the image of g spans T ; that is, $\text{span}(v \otimes w) = T$.

THEOREM A.1: (Uniqueness of Tensor Products) Let (T, g) and (T', g') be tensor products of V and W . Then there exists a unique isomorphism $h: T \rightarrow T'$ such that $hg = g'$.

Proof. Because T is a tensor product, and $g': V \times W \rightarrow T'$ is bilinear, there exists a unique linear map $h: T \rightarrow T'$ such that $hg = g'$. Similarly, because T' is a tensor product, and $g: V \times W \rightarrow T$ is bilinear, there exists a unique linear map $h': T' \rightarrow T$ such that $h'g' = g$. Using $hg = g'$, we get $h'hg = g$. Also, because T' is a tensor product, and $g: V \times W \rightarrow T$ is bilinear, there exists a unique linear map $h'': T' \rightarrow T$ such that $h''g' = g$. But $1_{T'}g' = g$. Thus, $h'h = h'' = 1_T$. Similarly, $hh' = 1_{T'}$. Therefore, h is an isomorphism from T to T' .

THEOREM A.2: (Existence of Tensor Product) The tensor product $T = V \otimes W$ of vector spaces V and W over K exists. Let $\{v_1, \dots, v_m\}$ be a basis of V and let $\{w_1, \dots, w_n\}$ be a basis of W . Then the mn vectors

$$v_i \otimes w_j \quad (i = 1, \dots, m; j = 1, \dots, n)$$

form a basis of T . Thus, $\dim T = mn = (\dim V) \dim (W)$:

Outline of Proof. Suppose $\{v_1, \dots, v_m\}$ is a basis of V , and suppose $\{w_1, \dots, w_n\}$ is a basis of W . Consider the mn

symbols $\{t_{ij} \mid i = 1; \dots; m; j = 1; \dots; n\}$. Let T be the vector space generated by the t_{ij} . That is, T consists of all linear combinations of the t_{ij} with coefficients in $\mathbf{K}$. [See [Problem 4.137.](#)]

Let $v \in V$ and $w \in W$. Say

$$v = a_1 v_1 + a_2 v_2 + \dots + a_m v_m \quad \text{and} \quad w = b_1 w_1 + b_2 w_2 + \dots + b_m w_m$$

Let $g: V \times W \rightarrow T$ be defined by

$$g(v, w) = \sum_i \sum_j a_i b_j t_{ij}$$

Then g is bilinear. [Proof left to reader.]

Now let $f: V \times W \rightarrow U$ be bilinear. Because the t_{ij} form a basis of T , [Theorem 5.2](#) (stated above) tells us that there exists a unique linear map $f^*: T \rightarrow U$ such that $f^* t_{ij} = f(v_i, w_j)$. Then, for $v = \sum_i a_i v_i$ and $w = \sum_j b_j w_j$ we have

$$f(v, w) = f\left(\sum_i a_i v_i, \sum_j b_j w_j\right) = \sum_i \sum_j a_i b_j f(v_i, w_j) = \sum_i \sum_j a_i b_j t_{ij} = f^*(g(v, w)).$$

Therefore, $f = f^* g$ where f^* is the required map in Definition A.1. Thus, T is a tensor product.

Let $\{v'_1, \dots, v'_m\}$ be any basis of V and $\{w'_1, \dots, w'_m\}$ be any basis of W .

Let $v \in V$ and $w \in W$ and say

$$v = a'_1 v'_1 + \dots + a'_m v'_m \quad \text{and} \quad w = b'_1 w'_1 + \dots + b'_m w'_m$$

Then

$$v \otimes w = g(v, w) = \sum_i \sum_j a'_i b'_j g(v'_i, w'_j) = \sum_i \sum_j a'_i b'_j (v'_i \otimes w'_j)$$

Thus, the elements $\underline{v'_i} \otimes \underline{w'_j}$ span T . There are mn such elements. They cannot be linearly dependent because $\{t_{ij}\}$ is a basis of T , and hence, $\dim T = mn$. Thus, the $\underline{v'_i} \otimes \underline{w'_j}$ form a basis of T .

Next we give two concrete examples of tensor products.

EXAMPLE A.1 Let V be the vector space of polynomials $P_{r-1}(x)$ and let W be the vector space of polynomials $P_{s-1}(y)$. Thus, the following form bases of V and W , respectively,

$$1, x, x^2, \dots, x^{r-1} \quad \text{and} \quad 1, y, y^2, \dots, y^{s-1}$$

In particular, $\dim V = r$ and $\dim W = s$: Let T be the vector space of polynomials in variables x and y with basis

$$\{x^i y^j\} \text{ where } i = 0, 1, \dots, r-1; j = 0, 1, \dots, s-1$$

Then T is the tensor product $V \otimes W$ under the mapping

$$x^i \otimes y^j = x^i y^j$$

For example, suppose $v = 2 \in 5x + 3x^3$ and $w = 7y + 4y^2$. Then

$$v \otimes w = 14y + 8y^2 - 35xy - 20xy^2 + 21x^3y + 12x^3y^2$$

EXAMPLE A.2

Let V be the vector space of $m \times n$ matrices over a field $\mathbf{K}$ and let W be the vector space of $p \times q$ matrices over $\mathbf{K}$. Suppose $A = [a_{11}]$ belongs to V , and B belongs to W . Let T be the vector space of $mp \times nq$ matrices over $\mathbf{K}$. Then T is the tensor product of V and W where $A \otimes B$ is the block matrix

$$A \otimes B = [a_{ij}B] = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix}$$

For example, suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$. Then

$$A \otimes B = \begin{bmatrix} 1 & 2 & 3 & 2 & 4 & 6 \\ 4 & 5 & 6 & 8 & 10 & 12 \\ 3 & 6 & 9 & 4 & 8 & 12 \\ 12 & 15 & 18 & 16 & 20 & 24 \end{bmatrix}$$

Isomorphisms of Tensor Products

First we note that tensoring is associative in a canonical way. Namely,

THEOREM A.3: Let U, V, W be vector spaces over a field $\mathbf{K}$. Then there exists a unique isomorphism

$$(U \otimes V) \otimes W \rightarrow U \otimes (V \otimes W)$$

such that, for every $u \in U; v \in V; w \in W$,

$$(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$$

Accordingly, we may omit parenthesis when tensoring any number of factors. Specifically, given vector spaces $V_1, V_2, \dots; V_m$ over a field $\mathbf{K}$, we may unambiguously form their tensor product

$$V_1 \otimes V_2 \otimes \dots \otimes V_m$$

and, for vectors v_j in V_j , we may unambiguously form the tensor product

$$v_1 \otimes v_2 \otimes \dots \otimes v_m$$

Moreover, given a vector space V over $\mathbf{K}$, we may unambiguously define the following tensor product:

$${}^rV = V \otimes V \dots \otimes V \text{ (} r \text{ factors)}$$

Also, there is a canonical isomorphism

$$(\otimes^r V) \otimes (\otimes^s V) \rightarrow \otimes^{r+s} V$$

Furthermore, viewing $\mathbf{K}$ as a vector space over itself, we have the canonical isomorphism

$$\mathbf{K} \otimes V \rightarrow V$$

where we define $a \otimes v = av$:

A.3 Alternating Multilinear Maps

Let $f: V^r \rightarrow U$ where V and U are vector spaces over K .
[Recall $V^r = V \times V \times \dots \times V$, r factors.]

- (1) The mapping f is said to be multilinear or r -linear if $f(v_1, \dots, v_r)$ is linear as a function of each v_j when the other v_i 's are held fixed. That is,

$$\begin{aligned} f(\dots, v_j + v'_j, \dots) &= f(\dots, v_j, \dots) + f(\dots, v'_j, \dots) \\ f(\dots, kv_j, \dots) &= kf(\dots, v_j, \dots) \end{aligned}$$

where only the j th position changes.

- (2) The mapping f is said to be alternating if

$$f(v_1, \dots, v_r) = 0 \text{ whenever } v_i = v_j \text{ with } i \neq j$$

One can easily show (Prove!) that if f is an alternating multilinear mapping on V^r , then

$$f(\dots, v_i, \dots, v_j, \dots) = -f(\dots, v_j, \dots, v_i, \dots)$$

That is, if two of the vectors are interchanged, then the associated value changes sign.

EXAMPLE A.3 (Determinants)

The determinant function $D: M \rightarrow K$ on the space M of $n \times n$ matrices may be viewed as an n -variable function

$$D(A) = D(R_1, R_2, \dots, R_n)$$

defined on the rows $R_1, R_2, \dots, R_n$ of A . Recall ([Chapter 8](#)) that, in this context, D is both n -linear and alternating.

We now need some additional notation. Let $K = k_1, k_2, \dots, k_r$ denote an r -list (r -tuple) of elements from $(I_n = 1; 2; \dots; n)$. We will then use the following notation where the v_k 's denote vectors and the a_{ik} 's denote scalars:

$$v_K = (v_{k_1}, v_{k_2}, \dots, v_{k_r}) \text{ and } a_K = a_{1k_1} a_{2k_2} \dots a_{rk_r}$$

Note v_K is a list of r vectors, and a_K is a product of r scalars.

Now suppose the elements in $K = k_1, k_2, \dots, k_r$ are distinct. Then K is a permutation s_K of an r -list $J = [i_1, i_2, \dots, i_r]$ in standard form, that is, where $i_1 < i_2 < \dots < i_r$. The number of such standard-form r -lists J from I_n is the binomial coefficient:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

[Recall $\text{sign}(\sigma_K) = (-1)^{m_K}$ where m_K is the number of interchanges that transforms K into J .]

Now suppose $A = a_{ij}$ is an $r \times n$ matrix. For a given ordered r -list J , we define

$$D_J(A) = \begin{vmatrix} a_{1i_1} & a_{1i_2} & \cdots & a_{1i_r} \\ a_{2i_1} & a_{2i_2} & \cdots & a_{2i_r} \\ \cdots & \cdots & \cdots & \cdots \\ a_{ri_1} & a_{ri_2} & \cdots & a_{ri_r} \end{vmatrix}$$

That is, $D_J(A)$ is the determinant of the $r \times r$ submatrix of A whose column subscripts belong to J .

Our main theorem below uses the following “shuffling” lemma.

LEMMA A.4 Let V and U be vector spaces over K , and let $f : V^r \rightarrow U$ be an alternating r -linear mapping. Let $v_1, v_2, \dots, v_n$ be vectors in V and let $A = a_{ij}$ be an $r \times n$ matrix over K where $r \geq n$. For $i = 1; 2; \dots; r$, let

$$u_i = a_{i1}v_1 + a_{i2}v_2 + \cdots + a_{in}v_n$$

Then

$$f(u_1, \dots, u_r) = \sum_f D_J(A) f(v_{i_1}, v_{i_2}, \dots, v_{i_r})$$

where the sum is over all standard-form r -lists $J = \{i_1, i_2, \dots, i_r\}$.

The proof is technical but straightforward. The linearity of f gives us the sum

$$f(u_1, \dots, u_r) = \sum_K a_K f(v_K)$$

where the sum is over all r -lists K from $\{1; \dots; n\}$. The alternating property of f tells us that $f(v_K) = 0$ when K does not contain distinct integers. The proof now mainly uses the fact that as we interchange the v_j 's to transform

$$f(v_K) = f(v_{k_1}, v_{k_2}, \dots, v_{k_r}) \quad \text{to} \quad f(v_j) = f(v_{i_1}, v_{i_2}, \dots, v_{i_r})$$

so that $i_1 < \dots < i_r$, the associated sign of a_K , will change in the same way as the sign of the corresponding permutation σ_K changes when it is transformed to the identity permutation using transpositions.

We illustrate the lemma below for $r = 2$ and $n = 3$.

EXAMPLE A.4 Suppose $f: V^2 \rightarrow U$ is an alternating multilinear function. Let $v_1, v_2, v_3 \in V$ and let $u, w \in V$. Suppose

$$u = a_1 v_1 + a_2 v_2 + a_3 v_3 \quad \text{and} \quad w = b_1 v_1 + b_2 v_2 + b_3 v_3$$

Consider

$$f(u, w) = f(a_1 v_1 + a_2 v_2 + a_3 v_3, b_1 v_1 + b_2 v_2 + b_3 v_3)$$

Using multilinearity, we get nine terms:

$$\begin{aligned} f(u, w) = & a_1 b_1 f(v_1, v_1) + a_1 b_2 f(v_1, v_2) + a_1 b_3 f(v_1, v_3) \\ & + a_2 b_1 f(v_2, v_1) + a_2 b_2 f(v_2, v_2) + a_2 b_3 f(v_2, v_3) \\ & + a_3 b_1 f(v_3, v_1) + a_3 b_2 f(v_3, v_2) + a_3 b_3 f(v_3, v_3) \end{aligned}$$

(Note that $J = [1; 2]$, $J' = [1; 3]$ and $J'' = 2; 3 \frac{1}{2}$ are the three standard-form 2-lists of $I = 1; 2; 3 \frac{1}{2}$.) The alternating property of f tells us that each $f(v_j, v_i) = 0$; hence, three of the above nine terms are equal to 0. The alternating property also tells us that $f(v_j, v_f) = f(v_f; v_r)$. Thus, three of the terms can be transformed so their subscripts form a standard-form 2-list by a single interchange. Finally we obtain

$$\begin{aligned} f(u, w) &= (a_1 b_2 - a_2 b_1) f(v_1, v_2) + (a_1 b_3 - a_3 b_1) f(v_1, v_3) + (a_2 b_3 - a_3 b_2) f(v_2, v_3) \\ &= \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} f(v_1, v_2) + \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} f(v_1, v_3) + \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} f(v_2, v_3) \end{aligned}$$

which is the content of Lemma A.4.

A.4 Exterior Products

The following definition applies.

DEFINITION A.2: Let V be an n -dimensional vector space over a field $\mathbf{K}$, and let r be an integer such that $1 \leq r \leq n$. The r -fold exterior product (or simply *exterior product* when r is understood) is a vector space $\mathbf{E}$ over $\mathbf{K}$ together with an alternating r -linear mapping $g: V^r \rightarrow \mathbf{E}$, denoted by $g(v_1, \dots, v_r) = v_1 \wedge \dots \wedge v_r$, with the following property: (*) For any vector space U over K and any alternating r -linear map $f: V^r \rightarrow U$ there exists a unique linear map $f^*: \mathbf{E} \rightarrow U$ such that $f^* \cdot g = f$.

The r -fold tensor product $(\mathbf{E}, g)$ (or simply $\mathbf{E}$ when g is understood) of V is denoted by $\Lambda^r V$, and the element $v_1 \wedge \dots$

$\wedge v_r$ is called the exterior product or wedge product of the v_i 's.

Another way to state condition (*) is that the diagram in Fig. A-1(c) commutes. Again, the fact that such a unique linear map f^* exists is called the "Universal Mapping Principle (UMP)". As illustrated in Fig. A-1 (c), condition (*) also says that any alternating r -linear map $f: V^r \rightarrow U$ "factors through" the exterior product $E = \wedge^r V$. Again, the uniqueness in (*) implies that the image of g spans E ; that is, $\text{span } v_1 \wedge \dots \wedge v_r = E$.

THEOREM A.5: (Uniqueness of Exterior Products) Let (E, g) and (E', g') be r -fold exterior products of V . Then there exists a unique isomorphism $h: E \rightarrow E'$ such that $hg = g'$.

The proof is the same as the proof of Theorem A.1, which uses the UMP.

THEOREM A.6: (Existence of Exterior Products) Let V be an n -dimensional vector space over K . Then the exterior product $E = \wedge^r V$ exists. If $r > n$, then $E = \{0\}$. If $r \geq n$, then $E = \binom{n}{r}$. Moreover, if $[v_1, \dots, v_n]$ is a basis of V , then the vectors

$$v_{i_1} \wedge v_{i_2} \wedge \dots \wedge v_{i_r},$$

where $1 \leq i_1 < i_2 < \dots < i_r \leq n$, form a basis of E .

We give a concrete example of an exterior product.

EXAMPLE A.5 (Cross Product)

Consider $V = \mathbf{R}^3$ with the usual basis $(\mathbf{i}, \mathbf{j}, \mathbf{k})$. Let $E = \wedge^2 V$. Note $\dim V = 3$: Thus, $\dim E = 3$ with basis $\mathbf{i} \wedge \mathbf{j}; \mathbf{i} \wedge \mathbf{k}; \mathbf{j} \wedge \mathbf{k}$

k: We identify E with $\mathbf{R}^3$ under the correspondence

$$\mathbf{i} = \mathbf{j} \wedge \mathbf{k}, \mathbf{j} = \mathbf{k} \wedge \mathbf{i} = -\mathbf{i} \wedge \mathbf{k}, \mathbf{k} = \mathbf{i} \wedge \mathbf{j}$$

Let u and w be arbitrary vectors in $V = \mathbf{R}^3$, say

$$u = (a_1, a_2, a_3) = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} \text{ and } w = (b_1, b_2, b_3) = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$$

Then, as in Example A.3,

$$u \wedge w = (a_1b_2 - a_2b_1)(\mathbf{i} \wedge \mathbf{j}) + (a_1b_3 - a_3b_1)(\mathbf{i} \wedge \mathbf{k}) + (a_2b_3 - a_3b_2)(\mathbf{j} \wedge \mathbf{k})$$

Using the above identification, we get

$$\begin{aligned} u \wedge w &= (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \\ &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} \end{aligned}$$

The reader may recognize that the above exterior product is precisely the well-known cross product in $\mathbf{R}^3$.

Our last theorem tells us that we are actually able to “multiply” exterior products, which allows us to form an “exterior algebra” that is illustrated below.

THEOREM A.7: Let V be a vector space over $\mathbf{K}$. Let r and s be positive integers. Then there is a unique bilinear mapping

$$\wedge^r V \times \wedge^s V \rightarrow \wedge^{r+s} V$$

such that, for any vectors u_i, w_j in V ,

$$(u_1 \wedge \cdots \wedge u_r) \times (w_1 \wedge \cdots \wedge w_s) \mapsto u_1 \wedge \cdots \wedge u_r \wedge w_1 \wedge \cdots \wedge w_s$$

EXAMPLE A.6 We form an exterior algebra A over a field K using noncommuting variables x, y, z . Because it is an

exterior algebra, our variables satisfy:

$$x \wedge x = 0, \quad y \wedge y = 0, \quad z \wedge z = 0, \quad \text{and} \quad y \wedge x = -x \wedge y, \quad z \wedge x = -x \wedge z, \quad z \wedge y = -y \wedge z$$

Every element of A is a linear combination of the eight elements 1;

$$1, \quad x, \quad y, \quad z, \quad x \wedge y, \quad x \wedge z, \quad y \wedge z, \quad x \wedge y \wedge z$$

We multiply two “polynomials” in A using the usual distributive law, but now we also use the above conditions. For example,

$$[3 + 4y - 5x \wedge y + 6x \wedge z] \wedge [5x - 2y] = 15x - 6y - 20x \wedge y + 12x \wedge y \wedge z$$

Observe we use the fact that

$$[4y] \wedge [5x] = 20y \wedge x = -20x \wedge y \quad \text{and} \quad [6x \wedge z] \wedge [-2y] = -12x \wedge z \wedge y = 12x \wedge y \wedge z$$

APPENDIX B

Algebraic Structures

B.1 Introduction

We define here algebraic structures that occur in almost all branches of mathematics. In particular, we will define a field that appears in the definition of a vector space. We begin with the definition of a group, which is a relatively simple algebraic structure with only one operation and is used as a building block for many other algebraic systems.

B.2 Groups

Let G be a nonempty set with a binary operation; that is, to each pair of elements $a; b \in G$ there is assigned an element $ab \in G$. Then G is called a *group* if the following axioms hold:

[G_1] For any $a; b; c \in G$, we have $(ab) c = a (bc)$ (the *associative law*).

[G_2] There exists an element $e \in G$, called the identity element, such that $ae = ea = a$ for every $a \in G$.

[G_3] For each $a \in G$ there exists an element $a^{-1} \in G$ called the inverse of a , such that $aa^{-1} = a^{-1}a = e$.

A group G is said to be abelian (or: commutative) if the commutative law holds—that is, if $ab = ba$ for every $a, b \in G$.

When the binary operation is denoted by juxtaposition as above, the group G is said to be written multiplicatively. Sometimes, when G is abelian, the binary operation is denoted by $+$ and G is said to be written additively. In such a case, the identity element is denoted by 0 and is called the zero element; the inverse is denoted by $-a$ and it is called the negative of a .

If A and B are subsets of a group G , then we write

$$AB = \{ab \mid a \in A, b \in B\} \quad \text{or} \quad A + B = \{a + b \mid a \in A, b \in B\}$$

We also write a for $\{a\}$.

A subset H of a group G is called a subgroup of G if H forms a group under the operation of G . If H is a subgroup of G and $a \in G$, then the set Ha is called a right coset of H and the set aH is called a left coset of H .

DEFINITION: A subgroup H of G is called a *normal* subgroup if $a^{-1}Ha \subseteq H$ for every $a \in G$. Equivalently, H is normal if $aH = Ha$ for every $a \in G$ —that is, if the right and left cosets of H coincide.

Note that every subgroup of an abelian group is normal.

THEOREM B.1: Let H be a normal subgroup of G . Then the cosets of H in G form a group under coset multiplication. This group is called the quotient group and is denoted by G/H .

EXAMPLE B.1 The set $\mathbf{Z}$ of integers forms an abelian group under addition. (We remark that the even integers form a subgroup of $\mathbf{Z}$ but the odd integers do not.) Let H denote the set of multiples of 5; that is, $H = \{\dots; 10; 5; 0; 5; 10; \dots\}$. Then H is a subgroup (necessarily normal) of $\mathbf{Z}$. The cosets of H in $\mathbf{Z}$ follow:

$$\bar{0} = 0 + H = H = \{\dots, -10, -5, 0, 5, 10, \dots\}$$

$$\bar{1} = 1 + H = \{\dots, -9, -4, 1, 6, 11, \dots\}$$

$$\bar{2} = 2 + H = \{\dots, -8, -3, 2, 7, 12, \dots\}$$

$$\bar{3} = 3 + H = \{\dots, -7, -2, 3, 8, 13, \dots\}$$

$$\bar{4} = 4 + H = \{\dots, -6, -1, 4, 9, 14, \dots\}$$

For any other integer $n \in \bar{n} = n + H$ coincides with one of the above cosets. Thus, by the above theorem,

$\mathbf{Z}/H = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ forms a group under coset addition; its addition table follows:

+	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{0}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{0}$	$\bar{1}$
$\bar{3}$	$\bar{3}$	$\bar{4}$	$\bar{0}$	$\bar{1}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$

This quotient group $\mathbf{Z}/H$ is referred to as the integers modulo 5 and is frequently denoted by $\mathbf{Z}_5$. Analogously, for any positive integer n , there exists the quotient group $\mathbf{Z}_n$ called the integers modulo n .

EXAMPLE B.2 The permutations of n symbols (see page 269) form a group under composition of mappings; it is

called the symmetric group of degree n and is denoted by S_n . We investigate S_3 here; its elements are

$$\begin{aligned} \epsilon &= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} & \sigma_2 &= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} & \phi_1 &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \\ \sigma_1 &= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} & \sigma_3 &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} & \phi_2 &= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \end{aligned}$$

Here $\begin{pmatrix} 1 & 2 & 3 \\ i & j & k \end{pmatrix}$ is the permutation that maps $1 \mapsto i$; $2 \mapsto j$; $3 \mapsto k$. The multiplication table of S_3 is

	ϵ	σ_1	σ_2	σ_3	ϕ_1	ϕ_2
ϵ	ϵ	σ_1	σ_2	σ_3	ϕ_1	ϕ_2
σ_1	σ_1	ϵ	ϕ_1	ϕ_2	σ_2	σ_3
σ_2	σ_2	ϕ_2	ϵ	ϕ_1	ϕ_3	σ_1
σ_3	σ_3	ϕ_1	ϕ_2	ϵ	σ_1	σ_2
ϕ_1	ϕ_1	σ_3	σ_1	σ_2	ϕ_2	ϵ
ϕ_2	ϕ_2	σ_2	σ_3	σ_1	ϵ	ϕ_1

(The element in the a th row and b th column is ab .) The set $H = \{\epsilon, \sigma_1\}$ is a subgroup of S_3 , its right and left cosets are

Right Cosets	Left Cosets
$H = \{\epsilon, \sigma_1\}$	$H = \{\epsilon, \sigma_1\}$
$H_{\phi_1} = \{\phi_1, \sigma_2\}$	$\phi_2 H = \{\phi_1, \sigma_3\}$
$H_{\phi_2} = \{\phi_2, \sigma_3\}$	$\phi_1 H = \{\phi_2, \sigma_2\}$

Observe that the right cosets and the left cosets are distinct; hence, H is not a normal subgroup of S_3 .

A mapping f from a group G into a group G is called a *homomorphism* if $f(ab) = f(a)f(b)$. For every $a, b \in G$. (If f is also bijective, i.e., one-to-one and onto, then f is called an isomorphism and G and G are said to be isomorphic.) If $f: G \rightarrow G$ is a homomorphism, then the kernel of f is the set of elements of G that map into the identity element $e' \in G$:
 kernel of $f = \{a \in G \mid f(a) = e'\}$

$$\text{kernel of } f = \{a \in G \mid f(a) = e'\}$$

(As usual, $f(G)$ is called the image of the mapping $f: G \rightarrow G$.) The following theorem applies.

THEOREM B.2: Let $f: G \rightarrow G$ be a homomorphism with kernel K . Then K is a normal subgroup of G , and the quotient group G/K is isomorphic to the image of f .

EXAMPLE B.3 Let G be the group of real numbers under addition, and let G be the group of positive real numbers under multiplication. The mapping $f: G \rightarrow G$ defined by $f(a) = 2^a$ is a homomorphism because

$$f(a + b) = 2^{a+b} = 2^a 2^b = f(a)f(b)$$

In particular, f is bijective, hence, G and G are isomorphic.

EXAMPLE B.4 Let G be the group of nonzero complex numbers under multiplication, and let G be the group of nonzero real numbers under multiplication. The mapping $f: G \rightarrow G$ defined by $f(a) = 2^a$ is a homomorphism because

$$f(z_1 z_2) = |z_1 z_2| = |z_1| |z_2| = f(z_1) f(z_2)$$

The kernel K of f consists of those complex numbers z on the unit circle—that is, for which $|z| = 1$. Thus, G/K is

isomorphic to the image of f —that is, to the group of positive real numbers under multiplication.

B.3 Rings, Integral Domains, and Fields

Let R be a nonempty set with two binary operations, an operation of addition (denoted by $+$) and an operation of multiplication (denoted by juxtaposition). Then R is called a ring if the following axioms are satisfied:

[R_1] For any $a; b; c \in R$, we have $(a + b) + c = a + (b + c)$.

[R_2] There exists an element $0 \in R$; called the zero element, such that $a + 0 = 0 + a = a$ for every $a \in R$.

[R_4] For each $a \in R$ there exists an element $-a \in R$, called the negative of a , such that $a + (-a) = (-a) + a = 0$.

[R_5] For any $a, b \in R$, we have $a + b = b + a$.

[R_5] For any $a, b, c \in R$ we have $(ab)c = a(bc)$.

[R_6] For any $a, b, c \in R$; we have

(i) $a(b + c) = ab + ac$, and (ii) $(b + c)a = ba + ca$.

Observe that the axioms [R_1] through [R_4] may be summarized by saying that R is an abelian group under addition.

Subtraction is defined in R by $a - b = a + (-b)$.

It can be shown (see Problem B.25) that $a \cdot 0 = 0 \cdot a = 0$ for every $a \in R$:

R is called a commutative ring if $ab = ba$ for every $a, b \in R$: We also say that R is a ring with a unit element if there exists a nonzero element $1 \in R$ such that $a \cdot 1 = 1 \cdot a = a$ for every $a \in R$:

A nonempty subset S of R is called a subring of R if S forms a ring under the operations of R . We note that S is a subring of R if and only if $a, b \in S$ implies $a - b \in S$ and $ab \in S$.

A nonempty subset I of R is called a left ideal in R if (i) $a - b \in I$ whenever $a, b \in I$; and (ii) $ra \in I$ whenever $r \in R$; $a \in I$: Note that a left ideal I in R is also a subring of R . Similarly, we can define a right ideal and a two-sided ideal. Clearly all ideals in commutative rings are two sided. The term ideal shall mean two-sided ideal unless otherwise specified.

THEOREM B.3: Let I be a (two-sided) ideal in a ring R . Then the cosets $\{a + I \mid a \in R\}$ form a ring under coset addition and coset multiplication. This ring is denoted by R/I and is called the quotient ring.

Now let R be a commutative ring with a unit element. For any $a \in R$, the set $(a) = \{ra \mid r \in R\}$ is an ideal; it is called the *principal* ideal generated by a . If every ideal in R is a principal ideal, then R is called a *principal ideal ring*.

DEFINITION: A commutative ring R with a unit element is called an integral domain if R has no zero divisors—that is, if $ab = 0$ implies $a = 0$ or $b = 0$.

DEFINITION: A commutative ring R with a unit element is called a *field* if every nonzero $a \in R$ has a multiplicative inverse; that is, there exists an element $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$.

A field is necessarily an integral domain; for if $ab = 0$ and $a \neq 0$; then

$$b = 1 \cdot b = a^{-1}ab = a^{-1} \cdot 0 = 0$$

We remark that a field may also be viewed as a commutative ring in which the nonzero elements form a group under multiplication.

EXAMPLE B.5 The set Z of integers with the usual operations of addition and multiplication is the classical example of an integral domain with a unit element. Every ideal I in Z is a principal ideal; that is, $I = (n)$ for some integer n . The quotient ring $\mathbf{Z}_n = \mathbf{Z}/I$ is called the *ring of integers module n* . If n is prime, then $\mathbf{Z}_n$ is a field. On the other hand, if n is not prime then $\mathbf{Z}_n$ has zero divisors. For example, in the ring $\mathbf{Z}_6$, $\bar{2}\bar{3} = \bar{0}$ and $\bar{2} \neq \bar{0}$ and $\bar{3} \neq \bar{0}$.

EXAMPLE B.6 The rational numbers Q and the real numbers R each form a field with respect to the usual operations of addition and multiplication.

EXAMPLE B.7 Let C denote the set of ordered pairs of real numbers with addition and multiplication defined by

$$\begin{aligned}(a, b) + (c, d) &= (a + c, b + d) \\ (a, b) \cdot (c, d) &= (ac - bd, ad + bc)\end{aligned}$$

Then C satisfies all the required properties of a field. In fact, C is just the field of complex numbers (see page 4).

EXAMPLE B.8 The set M of all 2×2 matrices with real entries forms a noncommutative ring with zero divisors under the operations of matrix addition and matrix multiplication.

EXAMPLE B.9 Let R be any ring. Then the set $R[x]$ of all polynomials over R forms a ring with respect to the usual

operations of addition and multiplication of polynomials. Moreover, if R is an integral domain then $R[x]$ is also an integral domain.

Now let D be an integral domain. We say that b divides a in D if $a = bc$ for some $c \in D$. An element $u \in D$ is called a *unit* if u divides 1—that is, if u has a multiplicative inverse. An element $b \in D$ is called an associate of $a \in D$ if $b = ua$ for some unit $u \in D$. A nonunit $p \in D$ is said to be *irreducible* if $p = ab$ implies a or b is a unit.

An integral domain D is called a *unique factorization domain* if every nonunit $a \in D$ can be written uniquely (up to associates and order) as a product of irreducible elements.

EXAMPLE B.10 The ring $\mathbf{Z}$ of integers is the classical example of a unique factorization domain. The units of $\mathbf{Z}$ are 1 and -1 . The only associates of $n \in \mathbf{Z}$ are n and $-n$. The irreducible elements of $\mathbf{Z}$ are the prime numbers.

EXAMPLE B.11 The set $D = \{a + b\sqrt{13} \mid a, b \text{ integers}\}$ is an integral domain. The units of D are ± 1 ; $18 \pm 5\sqrt{13}$ and $-18 \pm 5\sqrt{13}$. The elements 2 , $3 - \sqrt{13}$ and $-3 - \sqrt{13}$ are irreducible in D . Observe that $4 = 2 \cdot 2 = (3 - \sqrt{13})(-3 - \sqrt{13})$. Thus, D is not a unique factorization domain. (See Problem B.40.)

B.4 Modules

Let M be an additive abelian group and let R be a ring with a unit element. Then M is said to be a (left) R -*module* if there exists a mapping $R \rightarrow M \rightarrow M$ that satisfies the following axioms:

$$[M_1] \quad r(m_1 + m_2) = rm_1 + rm_2$$

$$[M_2] \quad (r + s)m = rm + sm$$

$$[M_3] \quad (rs)m = r(sm)$$

$$[M_4] \quad 1 \cdot m = m$$

for any $r, s \in R$ and any $m_i \in M$.

We emphasize that an R -module is a generalization of a vector space where we allow the scalars to come from a ring rather than a field.

EXAMPLE B.12 Let G be any additive abelian group. We make G into a module over the ring $\mathbf{Z}$ of integers by defining

$$ng = \overbrace{g + g + \cdots + g}^{n \text{ times}}, \quad 0g = 0, \quad (-n)g = -ng$$

where n is any positive integer.

EXAMPLE B.13 Let R be a ring and let I be an ideal in R . Then I may be viewed as a module over R .

EXAMPLE B.14 Let V be a vector space over a field K and let $T: V \rightarrow V$ be a linear mapping. We make V into a module over the ring $K[x]$ of polynomials over K by defining $f(x) \cdot v = f(T)(v)$: The reader should check that a scalar multiplication has been defined.

Let M be a module over R . An additive subgroup N of M is called a submodule of M if $u \in N$ and $k \in R$ imply $ku \in N$: (Note that N is then a module over R .)

Let M and M be R -modules. A mapping $T: M \rightarrow M$ is called a homomorphism (or: R -homomorphism or R -linear) if

$$(i) \quad T(u + v) = T(u) + T(v) \quad \text{and} \quad (ii) \quad T(ku) = kT(u)$$

for every $u, v \in M$ and every $k \in R$.

PROBLEMS

Groups

B.1. Determine whether each of the following systems forms a group G :

- (i) $G =$ set of integers; operation subtraction;
- (ii) $G = \{1, -1\}$, operation multiplication;
- (iii) $G =$ set of nonzero rational numbers, operation division;
- (iv) $G =$ set of nonsingular $n \times n$ matrices, operation matrix multiplication;
- (v) $G = \{a + bi : a, b \in \mathbf{Z}\}$, operation addition.

B.2. Show that in a group G :

- (i) the identity element of G is unique;
- (ii) each $a \in G$ has a unique inverse $a^{-1} \in G$;
- (iii) $(a^{-1})^{-1} = a$; and $(ab)^{-1} = b^{-1}a^{-1}$;
- (iv) $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.

B.3. In a group G , the powers of $a \in G$ are defined by

$$a^0 = e, \quad a^n = aa^{n-1}, \quad a^{-n} = (a^n)^{-1}, \quad \text{where } n \in \mathbf{N}$$

Show that the following formulas hold for any integers $r, s, t \in \mathbf{Z}$: (i) $a^r a^s = a^{r+s}$, (ii) $(a^r)^s = a^{rs}$

B.4. Show that if G is an abelian group, then $(ab)^n = a^n b^n$ for any $a, b \in G$ and any integer $n \in \mathbf{Z}$:

B.5. Suppose G is a group such that $(ab)^2 = a^2 b^2$ for every $a, b \in G$. Show that G is abelian.

B.6. Suppose H is a subset of a group G . Show that H is a subgroup of G if and only if (i) H is nonempty, and (ii) a ,

$b \in H$ implies $ab^{-1} \in H$.

- B.7.** Prove that the intersection of any number of subgroups of G is also a subgroup of G .
- B.8.** Show that the set of all powers of $a \in G$ is a subgroup of G ; it is called the cyclic group generated by a .
- B.9.** A group G is said to be cyclic if G is generated by some $a \in G$; that is, $G = \langle a \rangle : a \in G$. Show that every subgroup of a cyclic group is cyclic.
- B.10.** Suppose G is a cyclic subgroup. Show that G is isomorphic to the set $\mathbb{Z}$ of integers under addition or to the set $\mathbb{Z}_n$ (of the integers module n) under addition.
- B.11.** Let H be a subgroup of G . Show that the right (left) cosets of H partition G into mutually disjoint subsets.
- B.12.** The order of a group G , denoted by $|G|$; is the number of elements of G . Prove Lagrange's theorem: If H is a subgroup of a finite group G , then $|H|$ divides $|G|$.
- B.13.** Suppose $|G| = p$ where p is prime. Show that G is cyclic.
- B.14.** Suppose H and N are subgroups of G with N normal. Show that (i) HN is a subgroup of G and (ii) $H \cap N$ is a normal subgroup of G .
- B.15.** Let H be a subgroup of G with only two right (left) cosets. Show that H is a normal subgroup of G .
- B.16.** Prove Theorem B.1: Let H be a normal subgroup of G . Then the cosets of H in G form a group G/H under coset multiplication.
- B.17.** Suppose G is an abelian group. Show that any factor group G/H is also abelian.

- B.18.** Let $f: G \rightarrow G'$ be a group homomorphism. Show that
- (i) $f(e) = e'$ where e and e' are the identity elements of G and G' , respectively;
 - (ii) $f(a^{-1}) = f(a)^{-1}$ for any $a \in G$.
- B.19.** Prove Theorem B.2: Let $f: G \rightarrow G'$ be a group homomorphism with kernel K . Then K is a normal subgroup of G , and the quotient group G/K is isomorphic to the image of f .
- B.20.** Let G be the multiplicative group of complex numbers z such that $|z| = 1$; and let R be the additive group of real numbers. Prove that G is isomorphic to $\mathbf{R}/\mathbf{Z}$:
- B.21.** For a fixed $g \in G$, let $\hat{g}: G \rightarrow G$ be defined by $\hat{g}(a) = g^{-1}ag$. Show that $\hat{g}$ is an isomorphism of G onto G .
- B.22.** Let G be the multiplicative group of $n \times n$ nonsingular matrices over R . Show that the mapping $A \mapsto |A|$ is a homomorphism of G into the multiplicative group of nonzero real numbers.
- B.23.** Let G be an abelian group. For a fixed $n \in \mathbf{Z}$, show that the map $a \mapsto a^n$ is a homomorphism of G into G .
- B.24.** Suppose H and N are subgroups of G with N normal. Prove that $H \cap N$ is normal in H and $H/(H \cap N)$ is isomorphic to HN/N .

Rings

- B.25.** Show that in a ring R :
- (i) $a \cdot 0 = 0$ $a = 0$; (ii) $a(-b) = -(ab) = (-a)b$, (iii) $(-a)(-b) = ab$

- B.26.** Show that in a ring R with a unit element: (i) $(-1)a = a$, (ii) $(-1)(-1) = 1$.
- B.27.** Let R be a ring. Suppose $a^2 = a$ for every $a \in R$: Prove that R is a commutative ring. (Such a ring is called a *Boolean ring*.)
- B.28.** Let R be a ring with a unit element. We make R into another ring $\hat{R}$ by defining $a \oplus b = a + b + 1$ and $a \cdot b = ab + a + b$. (i) Verify that $\hat{R}$ is a ring. (ii) Determine the 0-element and 1-element of $\hat{R}$.
- B.29.** Let G be any (additive) abelian group. Define a multiplication in G by $a \cdot b = 0$. Show that this makes G into a ring.
- B.30.** Prove Theorem B.3: Let I be a (two-sided) ideal in a ring R . Then the cosets $(a + I \mid a \in R)$ form a ring under coset addition and coset multiplication.
- B.31.** Let I_1 and I_2 be ideals in R . Prove that $I_1 + I_2$ and $I_1 \cap I_2$ are also ideals in R .
- B.32.** Let R and R' be rings. A mapping $f : R \rightarrow R'$ is called a homomorphism (or: *ring homomorphism*) if
 (i) $f(a + b) = f(a) + f(b)$ and (ii) $f(ab) = f(a)f(b)$,
 for every $a, b \in R$. Prove that if $f : R \rightarrow R'$ is a homomorphism, then the set $K = \{r \in R \mid f(r) = 0\}$ is an ideal in R . (The set K is called the *kernel* of f .)

Integral Domains and Fields

- B.33.** Prove that in an integral domain D , if $ab = ac$; $a \neq 0$, then $b = c$.
- B.34.** Prove that $F = \{a + b\sqrt{2} \mid a, b \text{ rational}\}$ is a field.

- B.35.** Prove that $D = \{a + b\sqrt{2} \mid a, b \text{ integers}\}$ is an integral domain but not a field.
- B.36.** Prove that a finite integral domain D is a field.
- B.37.** Show that the only ideals in a field K are $\{0\}$ and K .
- B.38.** A complex number $a + bi$ where a, b are integers is called a Gaussian integer. Show that the set G of Gaussian integers is an integral domain. Also show that the units in G are ± 1 and $\pm i$.
- B.39.** Let D be an integral domain and let I be an ideal in D . Prove that the factor ring D/I is an integral domain if and only if I is a prime ideal. (An ideal I is prime if $ab \in I$ implies $a \in I$ or $b \in I$.)
- B.40.** Consider the integral domain $D = \{a + b\sqrt{13} \mid a, b \text{ integers}\}$ (see Example B.11). If $\alpha = a + b\sqrt{13}$ we define $N(\alpha) = a^2 - 13b^2$. Prove: (i) $N(\alpha\beta) = N(\alpha)N(\beta)$; (ii) α is a unit if and only if $N(\alpha) = \pm 1$; (iii) the units of D are $18 \pm 5\sqrt{13}$ and $-18 \pm 5\sqrt{13}$; (iv) the numbers $2, 3 - \sqrt{13}$ and $-3 - \sqrt{13}$ are irreducible.

Modules

- B.41.** Let M be an R -module and let A and B be submodules of M . Show that $A + B$ and $A \cap B$ are also submodules of M .
- B.42.** Let M be an R -module with submodule N . Show that the cosets $\{u + N : u \in M\}$ form an R -module under coset addition and scalar multiplication defined by $r(u + N) = ru + N$. (This module is denoted by M/N and is called the *quotient module*.)

- B.43.** Let M and M be R -modules and let $f: M \rightarrow M$ be an R -homomorphism. Show that the set $K = \{u \in M : f(u) = 0\}$ is a submodule of f . (The set K is called the kernel of f .)
- B.44.** Let M be an R -module and let $E(M)$ denote the set of all R -homomorphism of M into itself. Define the appropriate operations of addition and multiplication in $E(M)$ so that $E(M)$ becomes a ring.

APPENDIX C

Polynomials over a Field

C.1 Introduction

We will investigate polynomials over a field K and show that they have many properties that are analogous to properties of the integers. These results play an important role in obtaining canonical forms for a linear operator T on a vector space V over K .

C.2 Ring of Polynomials

Let K be a field. Formally, a polynomial of f over K is an infinite sequence of elements from K in which all except a finite number of them are 0:

$$f = (\dots, 0, a_n, \dots, a_1, a_0)$$

(We write the sequence so that it extends to the left instead of to the right.) The entry a_k is called the k th coefficient of f . If n is the largest integer for which $a_n \neq 0$, then we say that the degree of f is n , written

$$\deg f = n$$

We also call a_n the leading coefficient of f , and if $a_n = 1$ we call f a monic polynomial. On the other hand, if every coefficient of f is 0 then f is called the zero polynomial, written $f = 0$. The degree of the zero polynomial is not defined.

Now if g is another polynomial over K , say

$$g = (\dots, 0, b_m, \dots, b_1, b_0)$$

then the *sum* $f + g$ is the polynomial obtained by adding corresponding coefficients. That is, if $m \geq n$, then

$$f + g = (\dots, 0, a_n, \dots, a_m + b_m, \dots, a_1 + b_1, a_0 + b_0)$$

Furthermore, the product fg is the polynomial

$$fg = (\dots, 0, a_n b_m, \dots, a_1 b_0 + a_0 b_1, a_0 b_0)$$

that is, the k th coefficient c_k of fg is

$$c_k = \sum_{i=0}^k a_i b_{k-i} = a_0 b_k + a_1 b_{k-1} + \dots + a_k b_0$$

The following theorem applies.

THEOREM C.1: The set P of polynomials over a field K under the above operations of addition and multiplication forms a commutative ring with a unit element and with no zero divisors—an integral domain. If f and g are nonzero polynomials in P , then $\deg(fg) = (\deg f) + (\deg g)$.

Notation

We identify the scalar $a_0 \in K$ with the polynomial

$$a_0 = (\dots, 0, a_0)$$

We also choose a symbol, say t , to denote the polynomial

$$t = (\dots, 0, 1, 0)$$

We call the symbol t an indeterminate. Multiplying t with itself, we obtain

$$t^2 = (\dots, 0, 1, 0, 0), \quad t^3 = (\dots, 0, 1, 0, 0, 0), \quad \dots$$

Thus, the above polynomial f can be written uniquely in the usual form

$$f = a_n t^n + \dots + a_s t + a_0$$

When the symbol t is selected as the indeterminate, the ring of polynomials over K is denoted by

$$K[t]$$

and a polynomial f is frequently denoted by $f(t)$.

We also view the field K as a subset of $K[t]$ under the above identification. This is possible because the operations of addition and multiplication of elements of K are preserved under this identification:

$$\begin{aligned} (\dots, 0, a_0) + (\dots, 0, b_0) &= (\dots, 0, a_0 + b_0) \\ (\dots, 0, a_0) \cdot (\dots, 0, b_0) &= (\dots, 0, a_0 b_0) \end{aligned}$$

We remark that the nonzero elements of K are the units of the ring $K[t]$.

We also remark that every nonzero polynomial is an associate of a unique monic polynomial. Hence, if d and d' are monic polynomials for which d divides d' and d' divides d , then $d = d'$. (A polynomial g divides a polynomial f if there is a polynomial h such that $f = hg$.)

C.3 Divisibility

The following theorem formalizes the process known as “long division.”

THEOREM C.2 (Division Algorithm): Let f and g be polynomials over a field K with $g \neq 0$. Then there exist polynomials q and r such that

$$f = qg + r$$

where either $r = 0$ or $\deg r < \deg g$.

Proof: If $f = 0$ or if $\deg f < \deg g$, then we have the required representation

$$f = 0g + f$$

Now suppose $\deg f \geq \deg g$, say

$$f = a_n t^n + \cdots + a_1 t + a_0 \quad \text{and} \quad g = b_m t^m + \cdots + b_1 t + b_0$$

where $a_n, b_m \neq 0$ and $n \geq m$. We form the polynomial

$$f_1 = f - \frac{a_n}{b_m} t^{n-m} g$$

Then $\deg f_1 < \deg f$. By induction, there exist polynomials q_1 and r such that

$$f_1 = q_1 g + r$$

where either $r = 0$ or $\deg r < \deg g$. Substituting this into (1) and solving for f ,

$$f = \left(q_1 + \frac{a_n}{b_m} t^{n-m} \right) g + r$$

which is the desired representation.

THEOREM C.3: The ring $K[t]$ of polynomials over a field K is a principal ideal ring. If I is an ideal in $K[t]$, then there exists a unique monic polynomial d that generates I , such that d divides every polynomial $f \in I$.

Proof. Let d be a polynomial of lowest degree in I . Because we can multiply d by a nonzero scalar and still remain in I , we can assume without loss in generality that d is a monic polynomial. Now suppose $f \in I$. By Theorem C.2 there exist polynomials q and r such that

$$f = qd + r \text{ where either } r = 0 \text{ or } \deg r < \deg d$$

Now $f, d \in I$ implies $qd \in I$; and hence, $r = f - qd \in I$. But d is a polynomial of lowest degree in I . Accordingly, $r = 0$ and $f = qd$; that is, d divides f . It remains to show that d is unique. If d' is another monic polynomial that generates I , then d' divides d and d divides d' . This implies that $d = d'$, because d and d' are monic. Thus, the theorem is proved.

THEOREM C.4: Let f and g be nonzero polynomials in $K[t]$. Then there exists a unique monic polynomial d such that (i) d divides f and g ; and (ii) d divides f and g , then d divides d .

DEFINITION: The above polynomial d is called the greatest common divisor of f and g . If $d = 1$, then f and g are said to be relatively prime.

Proof of Theorem C.4. The set $I = \{mf + ng \mid m, n \in K[t]\}$ is an ideal. Let d be the monic polynomial that generates I . Note $f, g \in I$; hence, d divides f and g . Now suppose d divides f and g . Let J be the ideal generated by d . Then $f, g \in J$, and hence, $I \subset J$. Accordingly, $d \in J$ and so d divides d as claimed. It remains to show that d is unique. If d_1 is another (monic) greatest common divisor of f and g , then d divides d_1 and d_1 divides d . This implies that $d = d_1$ because d and d_1 are monic. Thus, the theorem is proved.

COROLLARY C.5: Let d be the greatest common divisor of the polynomials f and g . Then there exist polynomials m and n such that $d = mf + ng$. In particular, if f and g are relatively prime, then there exist polynomials m and n such that $mf + ng = 1$.

The corollary follows directly from the fact that d generates the ideal

$$I = \{mf + ng \mid m, n \in K[t]\}$$

C.4 Factorization

A polynomial $p \in K[t]$ of positive degree is said to be irreducible if $p = fg$ implies f or g is a scalar.

LEMMA C.6: Suppose $p \in K[t]$ is irreducible. If p divides the product fg of polynomials $f, g \in K[t]$, then p divides f or p divides g . More generally, if p divides the product of n

polynomials $f_1 f_2 \dots f_n$, then p divides one of them.

Proof. Suppose p divides fg but not f . Because p is irreducible, the polynomials f and p must then be relatively prime. Thus, there exist polynomials $m, n \in K[t]$ such that $mf + np = 1$. Multiplying this equation by g , we obtain $mfg + npg = g$. But p divides fg and so mfg , and p divides npg ; hence, p divides the sum $g = mfg + npg$.

Now suppose p divides $f_1 f_2 \dots f_n$: If p divides f_1 , then we are through. If not, then by the above result p divides the product $f_2 \dots f_n$: By induction on n , p divides one of the polynomials $f_2, \dots, f_n$: Thus, the lemma is proved.

THEOREM C.7: (Unique Factorization Theorem) Let f be a nonzero polynomial in $K[t]$: Then f can be written uniquely (except for order) as a product

$$f = kp_1 p_2 \cdots p_n$$

where $k \in K$ and the p_i are monic irreducible polynomials in $K[t]$.

Proof. We prove the existence of such a product first. If f is irreducible or if $f \in K$, then such a product clearly exists. On the other hand, suppose $f = gh$ where f and g are nonscalars. Then g and h have degrees less than that of f . By induction, we can assume

$$g = k_1 g_1 g_2 \cdots g_r \quad \text{and} \quad h = k_2 h_1 h_2 \cdots h_s$$

where $k_1, k_2 \in K$ and the g_i and h_j are monic irreducible polynomials. Accordingly,

$$f = (k_1 k_2) g_1 g_2 \cdots g_r h_1 h_2 \cdots h_s$$

is our desired representation.

We next prove uniqueness (except for order) of such a product for f . Suppose

$$f = kp_1p_2 \cdots p_n = k'q_1q_2 \cdots q_m$$

where $k, k' \in K$ and the $p_1, \dots, p_n, q_1, \dots, q_m$ are monic irreducible polynomials. Now p_1 divides $k'q_1 \cdots q_m$: Because p_1 is irreducible, it must divide one of the q_i by the above lemma. Say p_1 divides q_1 . Because p_1 and q_1 are both irreducible and monic, $p_1 = q_1$. Accordingly,

$$kp_2 \cdots p_n = k'q_2 \cdots q_m$$

By induction, we have that $n = m$ and $p_2 = q_2, \dots, p_n = q_m$ for some rearrangement of the q_i . We also have that $k = k'$. Thus, the theorem is proved.

If the field K is the complex field C , then we have the following result that is known as the fundamental theorem of algebra; its proof lies beyond the scope of this text.

THEOREM C.8: (Fundamental Theorem of Algebra) Let $f(t)$ be a nonzero polynomial over the complex field C . Then $f(t)$ can be written uniquely (except for order) as a product

$$f(t) = k(t - r_1)(t - r_2) \cdots (t - r_n)$$

where $k, r_i \in C$ —as a product of linear polynomials.

In the case of the real field **R** we have the following result.

THEOREM C.9: Let $f(t)$ be a nonzero polynomial over the real field **R**. Then $f(t)$ can be written uniquely

(except for order) as a product

$$f(t) = kp_1(t)p_2(t) \cdots p_m(t)$$

where $k \in \mathbf{R}$ and the $p_i(t)$ are monic
irreducible polynomials of degree one or two

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APPENDIX D

Odds and Ends

D.1 Introduction

This appendix discusses various topics, such as equivalence relations and singular value decomposition.

D.2 Relations and Equivalence Relations

A binary relation or simply *relation* R from a set A to a set B assigns to each ordered pair $(a, b) \in A \times B$ exactly one of the following statements:

(i) “ a is related to b ,” written $a R b$, (ii) “ a is not related to b ” written $a \not R b$.

A relation from a set A to the same set A is called a relation *on* A .

Observe that any relation R from A to B uniquely defines a subset $\hat{R}$ of $A \times B$ as follows:

$$\hat{R} = \{(a, b) | a R b\}$$

Conversely, any subset $\hat{R}$ of $A \times B$ defines a relation from A to B as follows:

$$a R b \text{ if and only if } (a, b) \in R$$

In view of the above correspondence between relations from A to B and subsets of $A \times B$, we redefine a relation from A to B as follows:

DEFINITION D.1: A relation R from A to B is a subset of $A \times B$.

Equivalence Relations

Consider a nonempty set S . A relation R on S is called an equivalence relation if R is reflexive, symmetric, and transitive; that is, if R satisfied the following three axioms:

[E₁] (Reflexivity) Every $a \in A$ is related to itself. That is, for every $a \in A$, $a R a$.

[E₂] (Symmetry) If a is related to b , then b is related to a . That is,

if $a R b$, then $b R a$.

[E₃] (Transitivity) If a is related to b and b is related to c , then a is related to c . That is, if $a R b$ and $b R c$, then $a R c$:

The general idea behind an equivalence relation is that it is a classification of objects that are in some way “alike.” Clearly, the relation of equality is an equivalence relation. For this reason, one frequently uses $\sim$ to denote an equivalence relation.

EXAMPLE D.1

(a) In Euclidean geometry, similarity of triangles is an equivalence relation. Specifically, suppose α , β , γ are

triangles. Then (i) α is similar to itself. (ii). If α is similar to β , then β is similar to α . (iii) If α is similar to β and β is similar to γ , then α is similar to γ .

(b) The relation $\subseteq$ of set inclusion is not an equivalence relation. It is reflexive and transitive, but it is not symmetric because $A \subseteq B$ does not imply $B \subseteq A$.

Equivalence Relations and Partitions

Let S be a nonempty set. Recall first that a partition P of S is a subdivision of S into nonempty, nonoverlapping subsets; that is, a collection $P = \{A_j\}$ of nonempty subsets of S such that (i) Each $a \in S$ belong to one of the A_j , (ii) The sets $\{A_j\}$ are mutually disjoint.

The subsets in a partition P are called cells. Thus, each $a \in S$ belongs to exactly one of the cells. Also, any element $b \in A_j$ is called a representative of the cell A_j , and a subset B of S is called a system of representatives if B contains exactly one element in each of the cells in $\{A_j\}$.

Now suppose R is an equivalence relation on the nonempty set S . For each $a \in S$, the equivalence class of a , denoted by $[a]$, is the set of elements of S to which a is related:

$$[a] = \{x \mid aRx\}.$$

The collection of equivalence classes, denoted by S/R , is called the quotient of S by R :

$$S/R = \{[a] \mid a \in S\}$$

The fundamental property of an equivalence relation and its quotient set is contained in the following theorem:

THEOREM D.1: Let R be an equivalence relation on a nonempty set S . Then the quotient set S/R is

a partition of S .

EXAMPLE D.2 Let $\equiv$ be the relation on the set $\mathbf{Z}$ of integers defined by $x \equiv y \pmod{5}$ which reads “ x is congruent to y modulus 5” and which means that the difference $x - y$ is divisible by 5. Then $\equiv$ is an equivalence relation on $\mathbf{Z}$.

Then there are exactly five equivalence classes in the quotient set $\mathbf{Z}/\equiv$ as follows:

$$A_0 = \{\dots, -10, -5, 0, 5, 10, \dots\}$$

$$A_1 = \{\dots, -9, -4, 1, 6, 11, \dots\}$$

$$A_2 = \{\dots, -8, -3, 2, 7, 12, \dots\}$$

$$A_3 = \{\dots, -7, -2, 3, 8, 13, \dots\}$$

$$A_4 = \{\dots, -6, -1, 4, 9, 14, \dots\}$$

Note that any integer x , which can be expressed uniquely in the form $x = 5q + r$ where $0 \leq r < 5$, is a member of the equivalence class A_r where r is the remainder. As expected, the equivalence classes are disjoint and their union is $\mathbf{Z}$:

$$\mathbf{Z} = A_0 \cup A_1 \cup A_2 \cup A_3 \cup A_4$$

This quotient set $\mathbf{Z}/\equiv$, called the *integers modulo 5*, is denoted

$\mathbf{Z}/5\mathbf{Z}$ or simply **$\mathbf{Z}_5$** :

Usually one chooses $\{0, 1, 2, 3, 4\}$ or $\{-2, -1, 0, 1, 2\}$ as a system of representatives of the equivalence classes.

Analagously, for any positive integer m , there exists the congruence relation defined by

$$x \equiv y \pmod{m}$$

and the quotient set $\mathbf{Z}/\equiv$, denoted by $\mathbf{Z}/m\mathbf{Z}$ or simply $\mathbf{Z}_m$, is called the *integers modulo m*.

D.3 Properties of AA^T and A^TA

Let A be any $m \times n$ matrix. Then A^TA and AA^T are both symmetric since

$$(A^TA)^T = A^T A^{TT} = A^TA \quad \text{and} \quad (AA^T)^T = A^{TT} A^T = AA^T$$

One can also show that A^TA and AA^T have the same nonzero eigenvalues.

Recall [Theorem 9.14] that a symmetric matrix M is orthogonally diagonalizable, that is, there exists an orthogonal matrix P and diagonal matrix $D = [d_i]$ such that

$$P^{-1}MP = D$$

where the columns of P are the eigenvectors of M and the d_i are the eigenvalues of M .

In particular, the symmetric matrix M is called:

- (i) *positive definite* if, for every nonzero u , $\langle u, Mu \rangle = u^T Mu > 0$
- (ii) *positive semidefinite* if, for every nonzero u , $\langle u, Mu \rangle = u^T Mu \geq 0$

In such a case, the diagonal elements in $P^{-1}MP = D$ are: (i) positive, (ii) nonnegative.

A^TA is positive definite if A has full column rank since $Au \neq 0$ when $u \neq 0$; so

$$\langle u, A^TAu \rangle = u^T A^TAu = \langle Au, Au \rangle > 0$$

Similarly, AA^T is positive definite if A has full row rank. since $A^T u \neq 0$ when $u \neq 0$; so

$$\langle u, AA^T u \rangle = u^T AA^T u = u^T A^T A^T A^T u = \langle A^T u, A^T u \rangle > 0$$

On the other hand, $A^T A$ and AA^T are always both positive semidefinite.

D.4 Singular Value Decomposition

Let A be any $m \times n$ matrix of rank r . Then there exists a factorization of A of the form

$$A = U \Sigma V^T$$

where U is an m -square orthogonal matrix, V is an n -square orthogonal matrix, and $\Sigma = [\sigma_{ij}]$ is an $m \times n$ generalized diagonal matrix, that is, $\sigma_{ij} = 0$ for $i \neq j$. Such a factorization is called a **singular value decomposition (SVD) of A** , and the diagonal entries $\sigma_i = \sigma_{ii}$ of Σ , usually listed in descending order, are called the *singular values* of A .

THEOREM D.2: Every matrix A with rank $r > 0$ has a **singular value decomposition**.

We indicate how the entries U , V and Σ in the SVD of A are obtained.

Recall AA^T is symmetric and positive definite (or positive semidefinite). Accordingly, AA^T is orthogonally diagonalizable, that is, $AA^T = P^{-1}DP = P^TDP$ where the columns of the orthogonal matrix P are eigenvectors of AA^T , and the entries of D are the eigenvalues of AA^T and they are nonnegative.

Assuming $A = U \Sigma V^T$, we have

$$A A^T = U \Sigma V^T (U \Sigma V^T)^T = U \Sigma V^T V \Sigma U^T = U \Sigma^2 U^T$$

Thus the columns of U in the SVD of A are the normalized eigenvectors of AA^T and the entries σ_i in Σ are the square roots of the eigenvalues of $A^T A$.

Similarly, assuming $A = U \Sigma V^T$, we have

$$A^T A = (U \Sigma V^T)^T U \Sigma V^T = V \Sigma U^T U \Sigma V^T = V \Sigma^2 V^T$$

Thus the columns of V and the rows of V^T are the normalized eigenvectors of $A^T A$ and the entries s_i in S are the square roots of the eigenvalues of AA^T (which are the same as the eigenvalues of $A^T A$).

EXAMPLE D.3. We find the singular value decomposition

(SVD) of $A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$. Note

$$AA^T = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}$$

The eigenvalues are 32 and 18 with corresponding eigenvectors $[1, 0]^T$ and $[0, 1]^T$. Thus

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } \Sigma = \begin{bmatrix} \sqrt{32} & 0 \\ 0 & \sqrt{18} \end{bmatrix} = \sqrt{2} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$$

Also

$$A^T A = \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 25 & 7 \\ 7 & 25 \end{bmatrix}$$

Hence $\Delta(t) = t^2 - 50t + 576 = (t - 32)(t - 18)$. Thus, again, the eigenvalues are 32 and 18.

The normalized corresponding eigenvectors are $[1/\sqrt{2}, 1/\sqrt{2}]^T$ and $[-1/\sqrt{2}, 1/\sqrt{2}]^T$. Thus

$$V^T = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

As expected,

$$U \Sigma V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \sqrt{2} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = A$$

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